

R2P Program Unit 2 - Lesson 1

Variational Algorithms

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Overview

Motivation:

- Current challenges of Quantum Computing
- Why should we use Variational Algorithms?

Variational Algorithms

- Concept
- Ansatz
- Reference State
- Cost Functions
- Optimization Loops

QAOA: Variational algorithms for optimization

- QUBO
- Adiabatic theorem
- Workflow

VQE: Variational algorithms for ground energy estimation

- Variational principle
- Workflow

ibm_torino

OpenQASM 3

Details

133

Qubits

0.8%

EPLG

3.8K

CLOPS

Status: ● Online

Total pending jobs: 43 jobs

Processor type ⓘ: Heron r1

Version: 1.0.16

Basis gates: CZ, ID, RZ, SX, X

Your instance usage: 0 jobs

Median CZ error: 4.384e-3

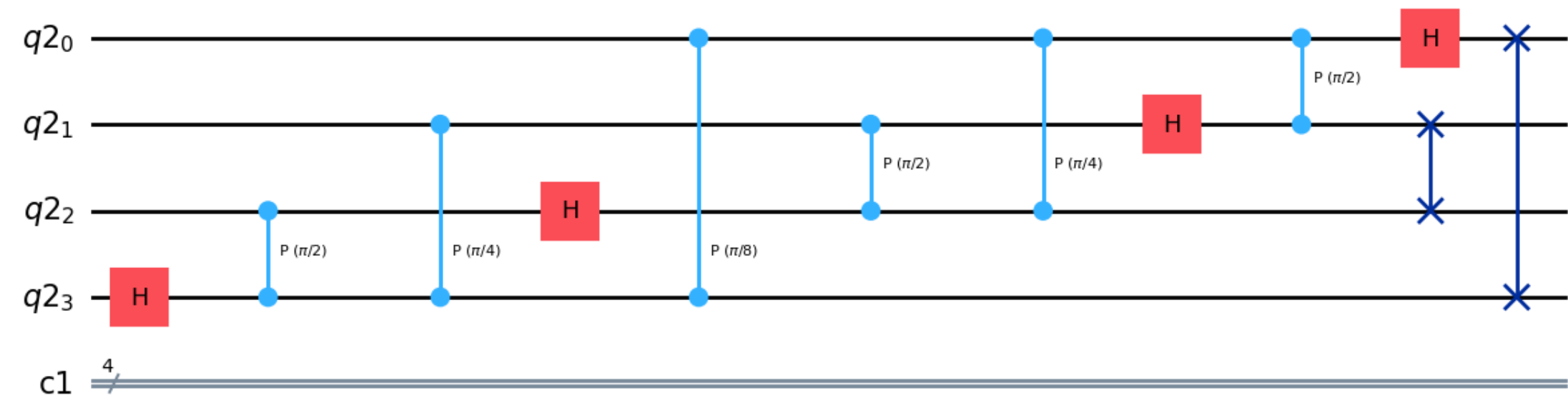
Median SX error: 3.161e-4

Median readout error: 1.770e-2

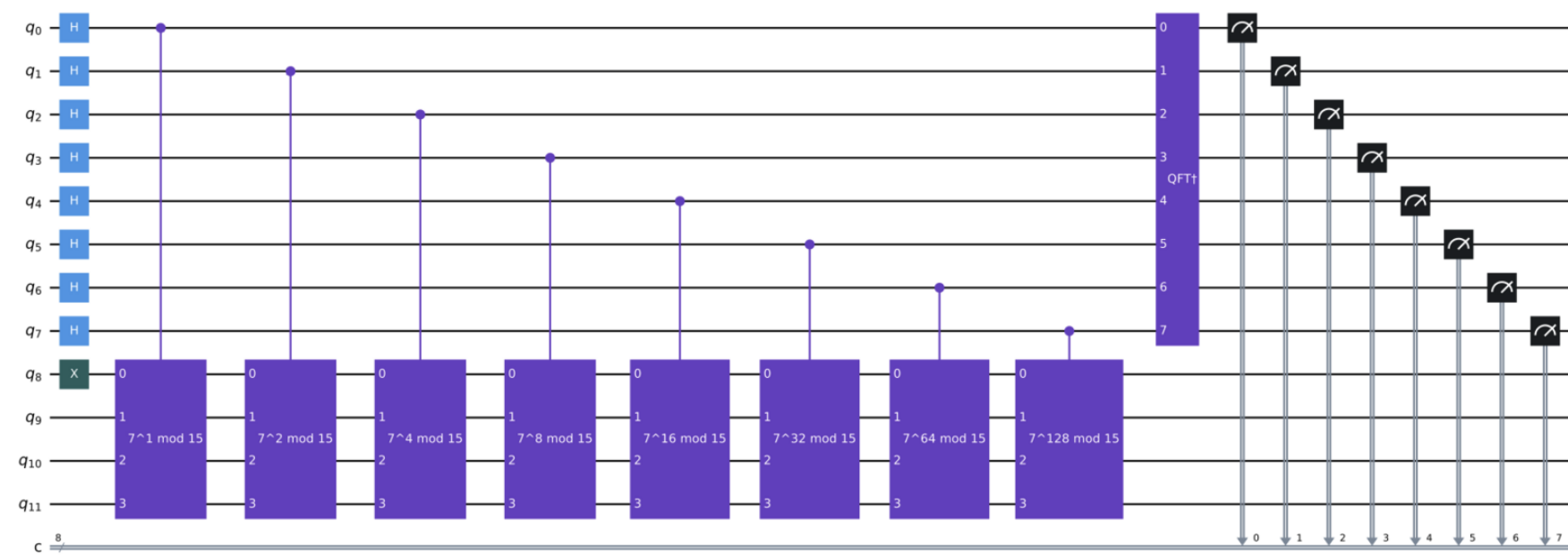
Median T1: 175.89 us

Median T2: 134.83 us

Noise in various forms.
Including noise in gates.

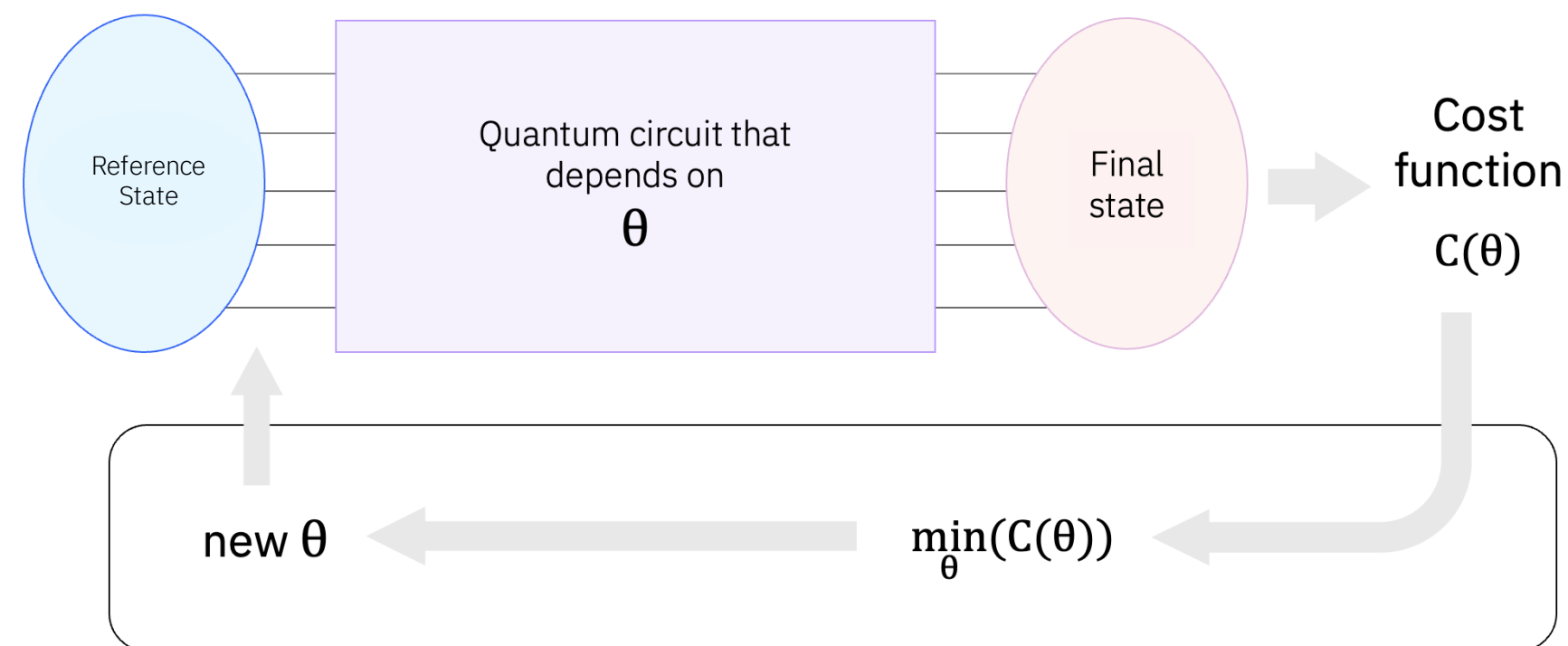
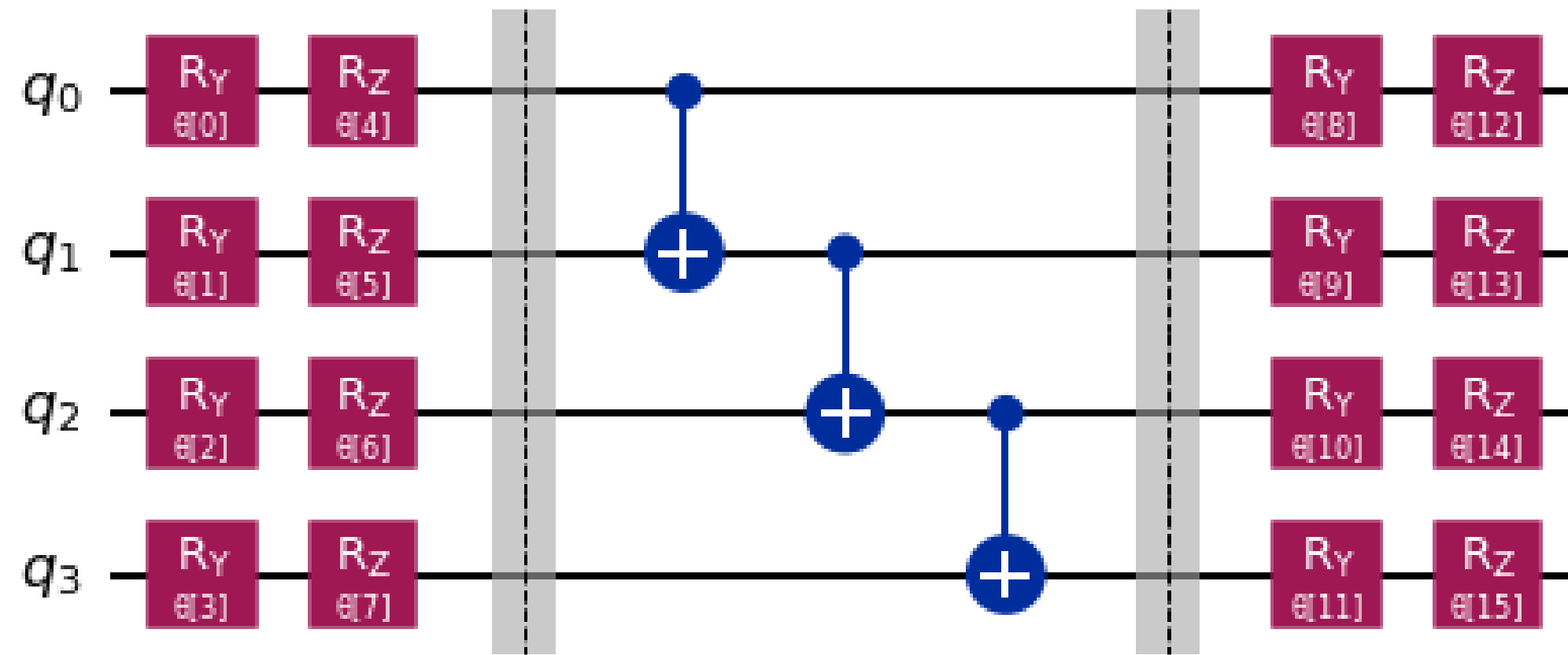


Depth is limited because of noise,
both decoherence, but also gate
noise.



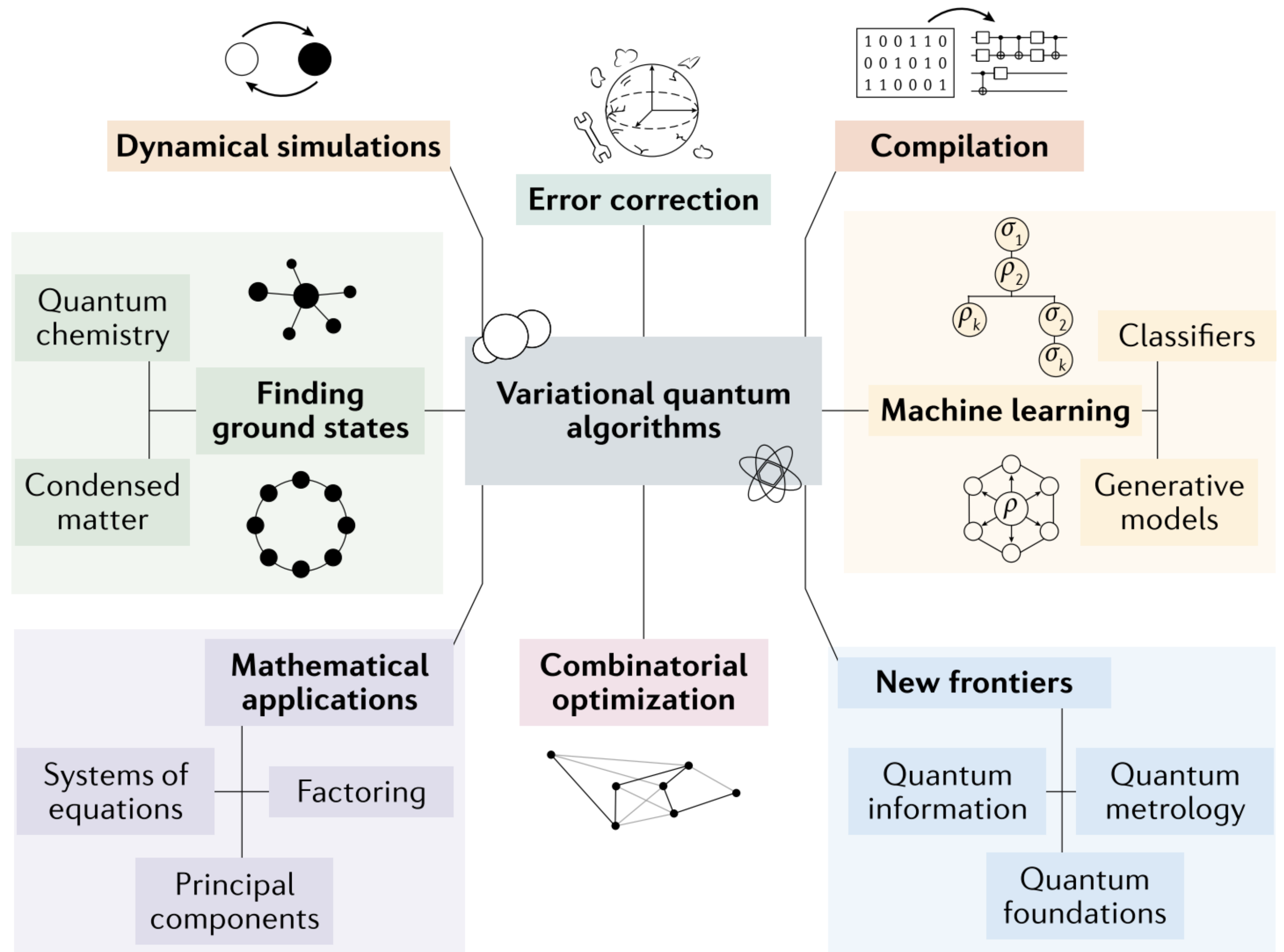
Measurements needed to get high
precision grow quickly with the
number of qubits.

Why should we use Variational Algorithms?

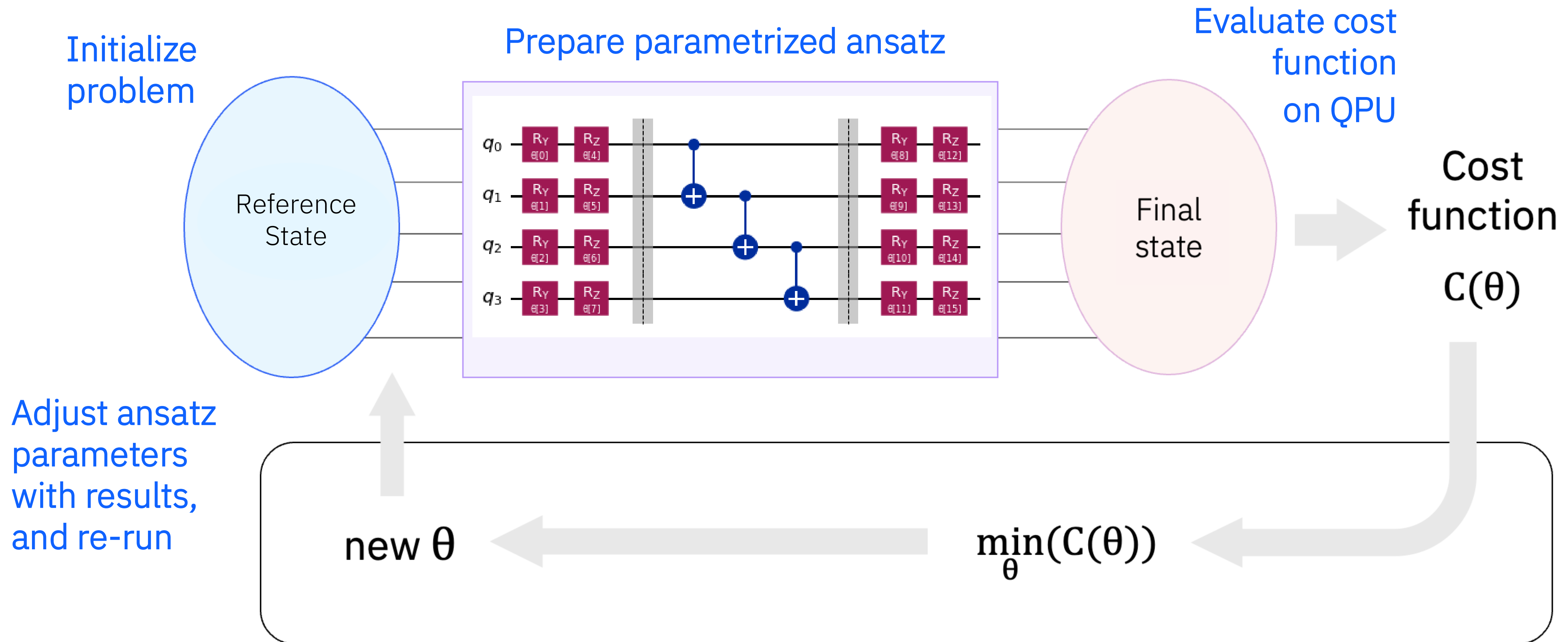


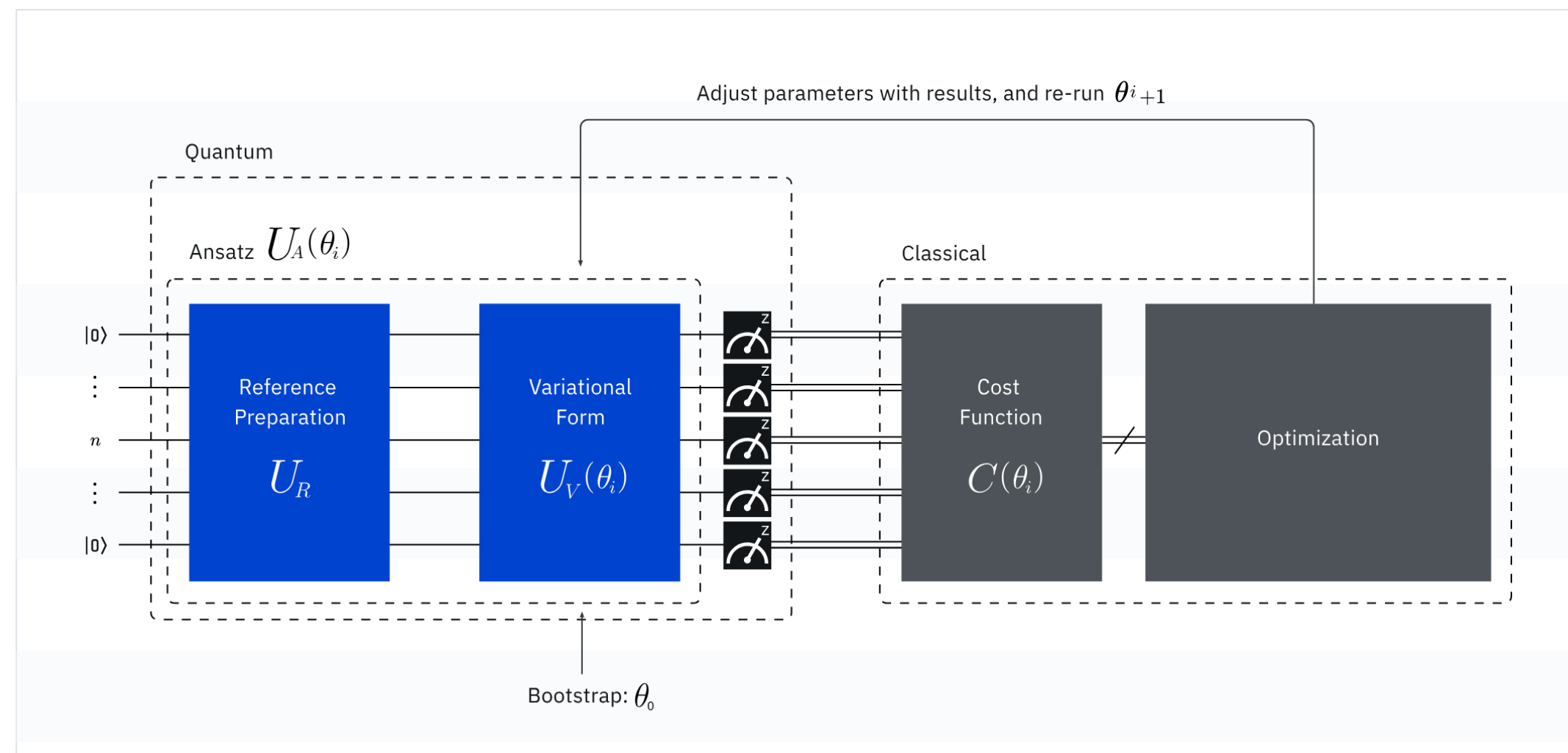
- They can be implemented with relatively low circuit depth.
- Less sensitive to noise due to their variational nature
- Can be combined with classical optimization allowing for hybrid algorithms.
- We are often interested in a simple expectation value and can use various forms of error mitigation.

Variational Algorithms

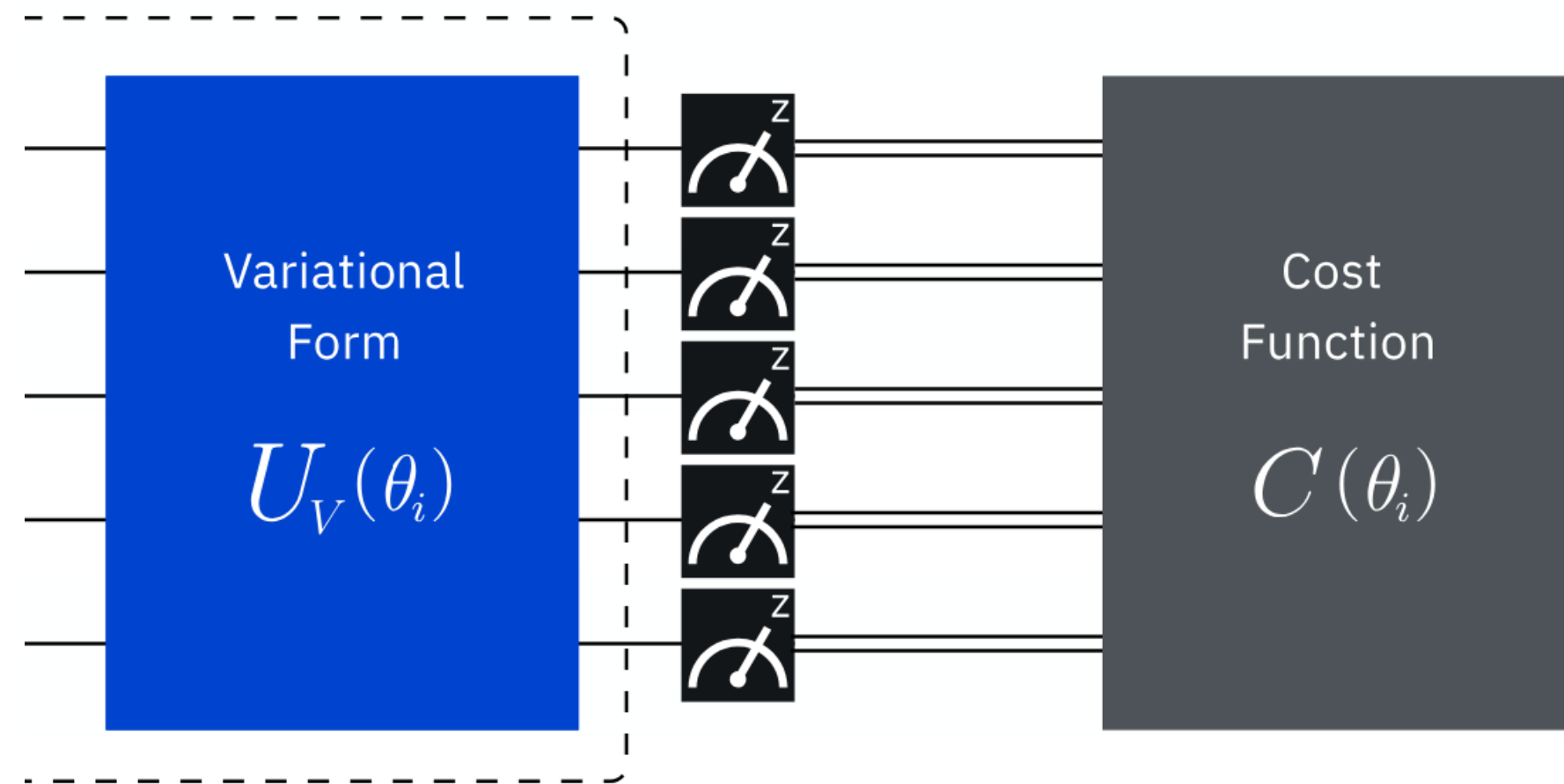


Cerezo, et al. Nature Reviews Physics 3, 625-644 (2021)

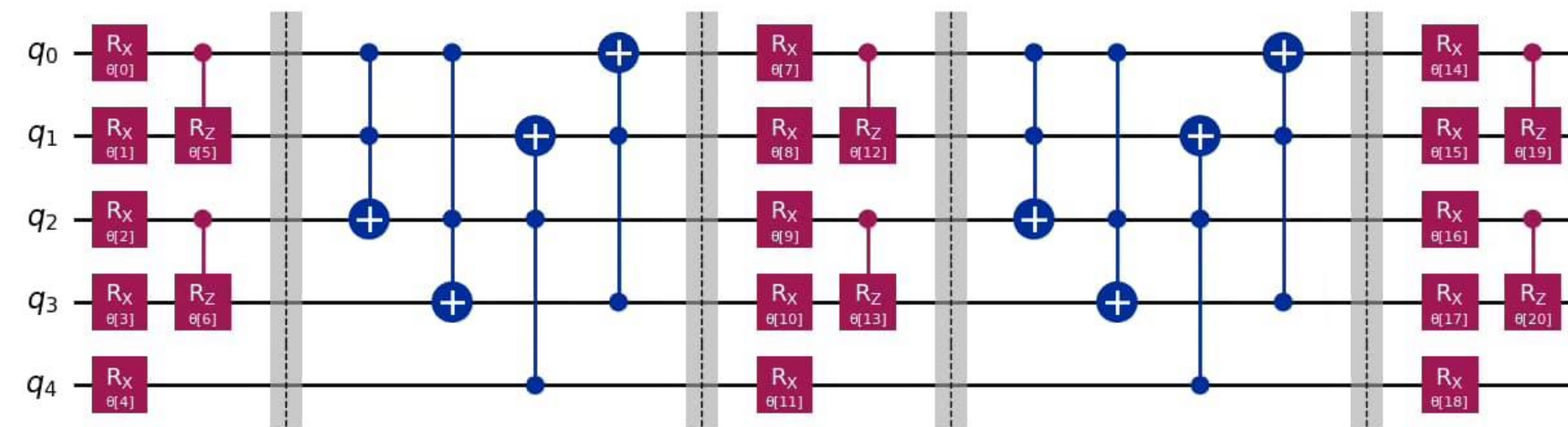




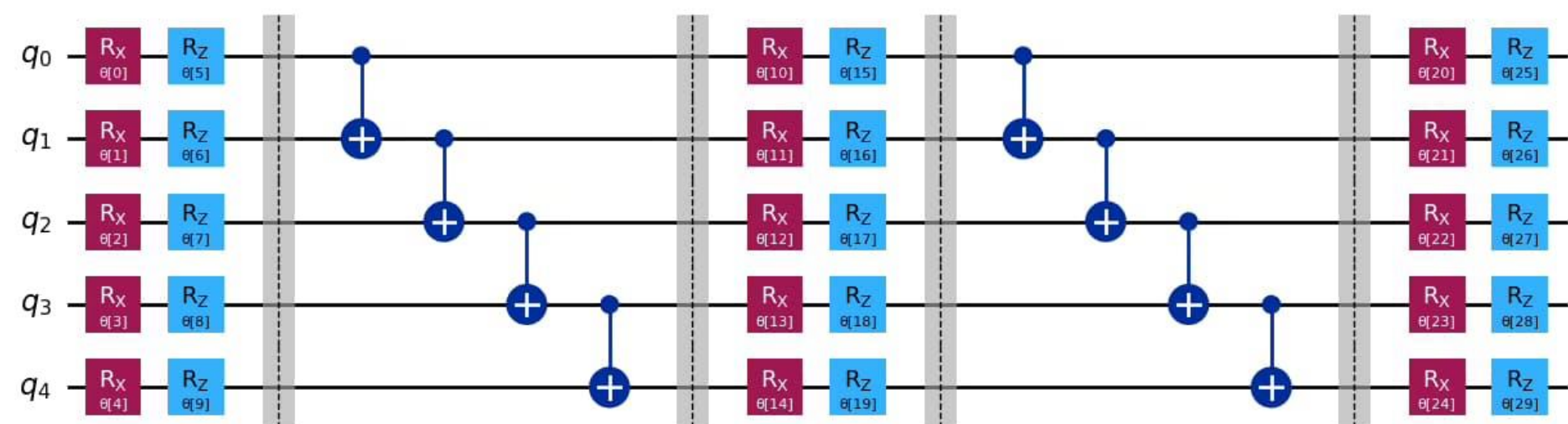
1. Define the quantity you want to minimize
2. Find an Ansatz to model the problem
3. Find a good initial state
4. Run the classical optimization loop
5. Experiment with different settings



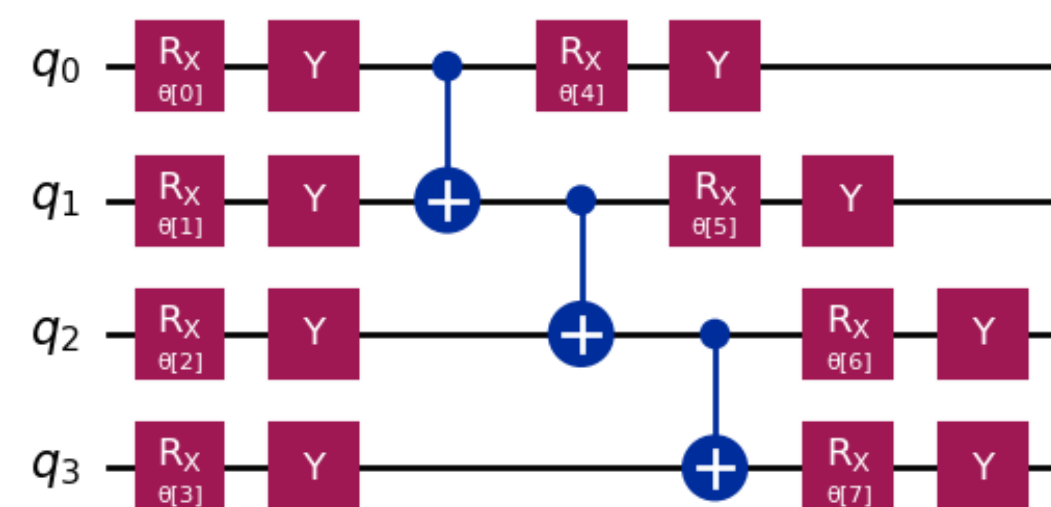
- The cost function defines the quantity we are trying to minimize. (e.g. Hamiltonian, the observable which defines the energy.)
- Can in general be any function acting on any subset of qubits.



n_local



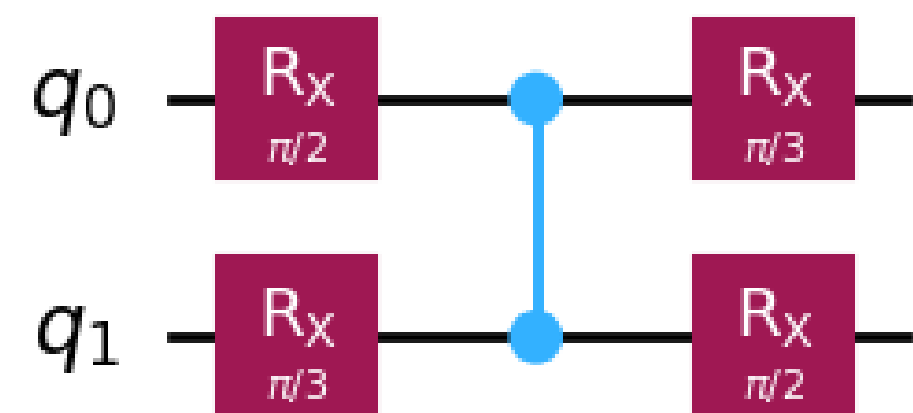
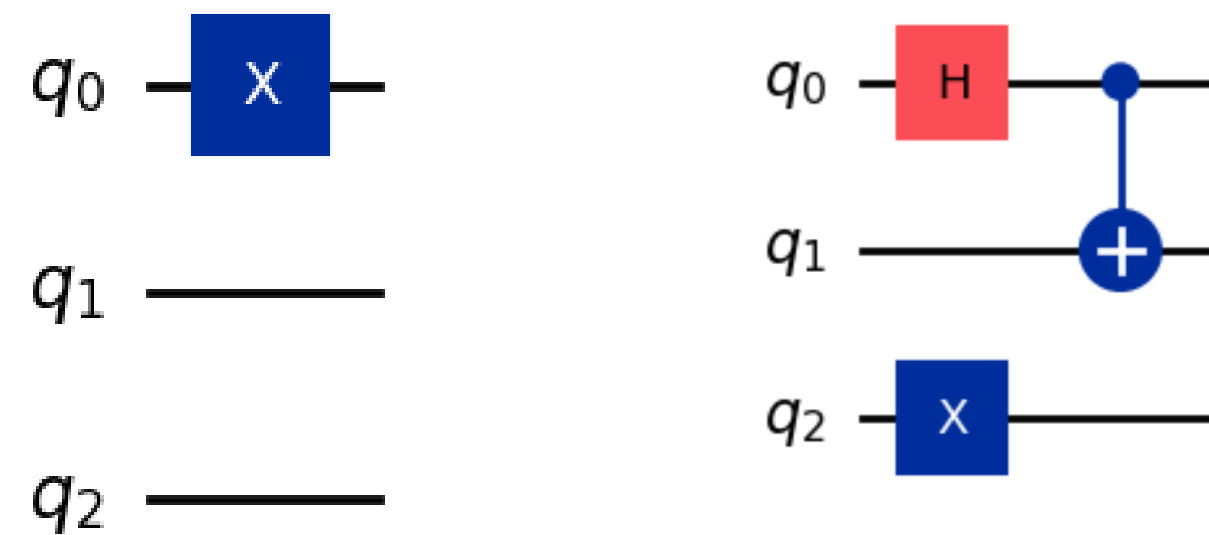
TwoLocal



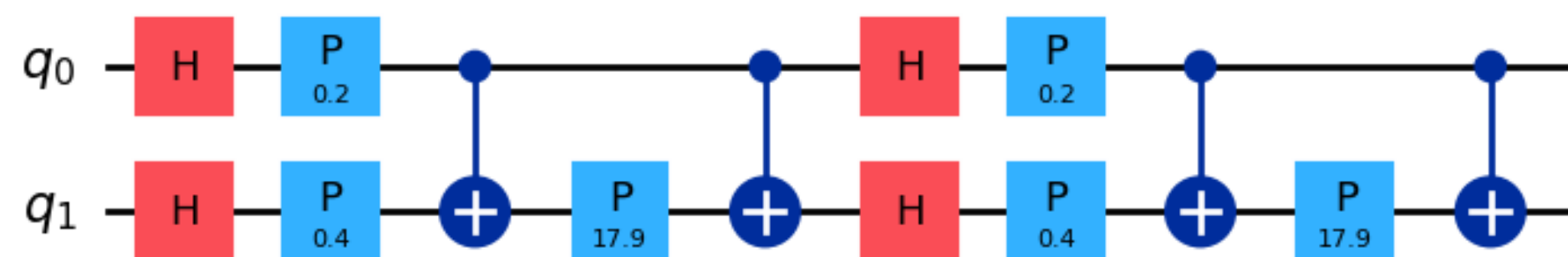
efficient_su2

- An Ansatz is a parameterized quantum circuit that prepares a quantum state which depends on a set of tuneable parameters.
- The goal is to reach a target state given the Ansatz and an initial state.
- Using knowledge about the result to lower the dimension space (parameters) to increase the speed and the chances for convergence.

Reference State



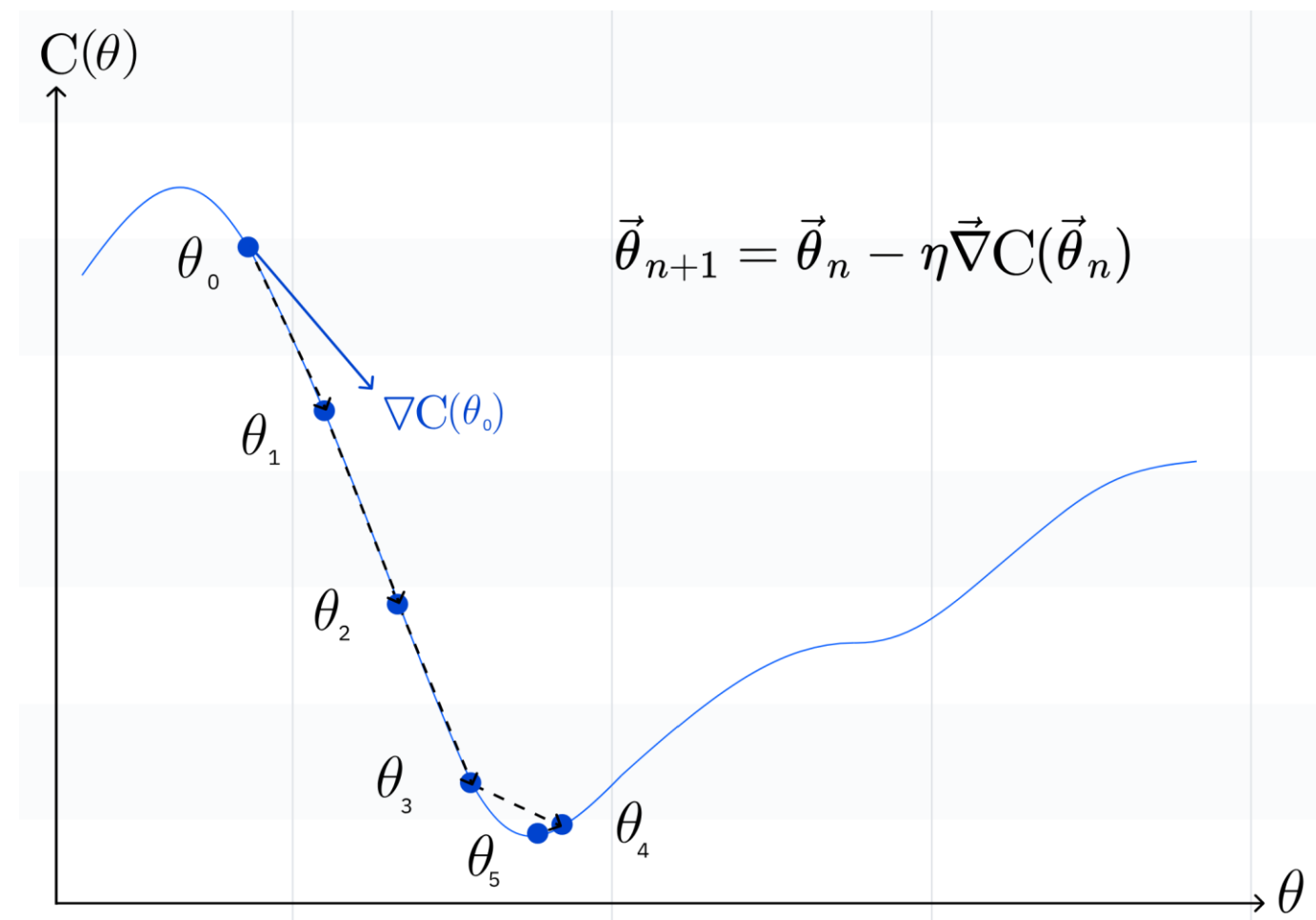
[TwoLocal](#)



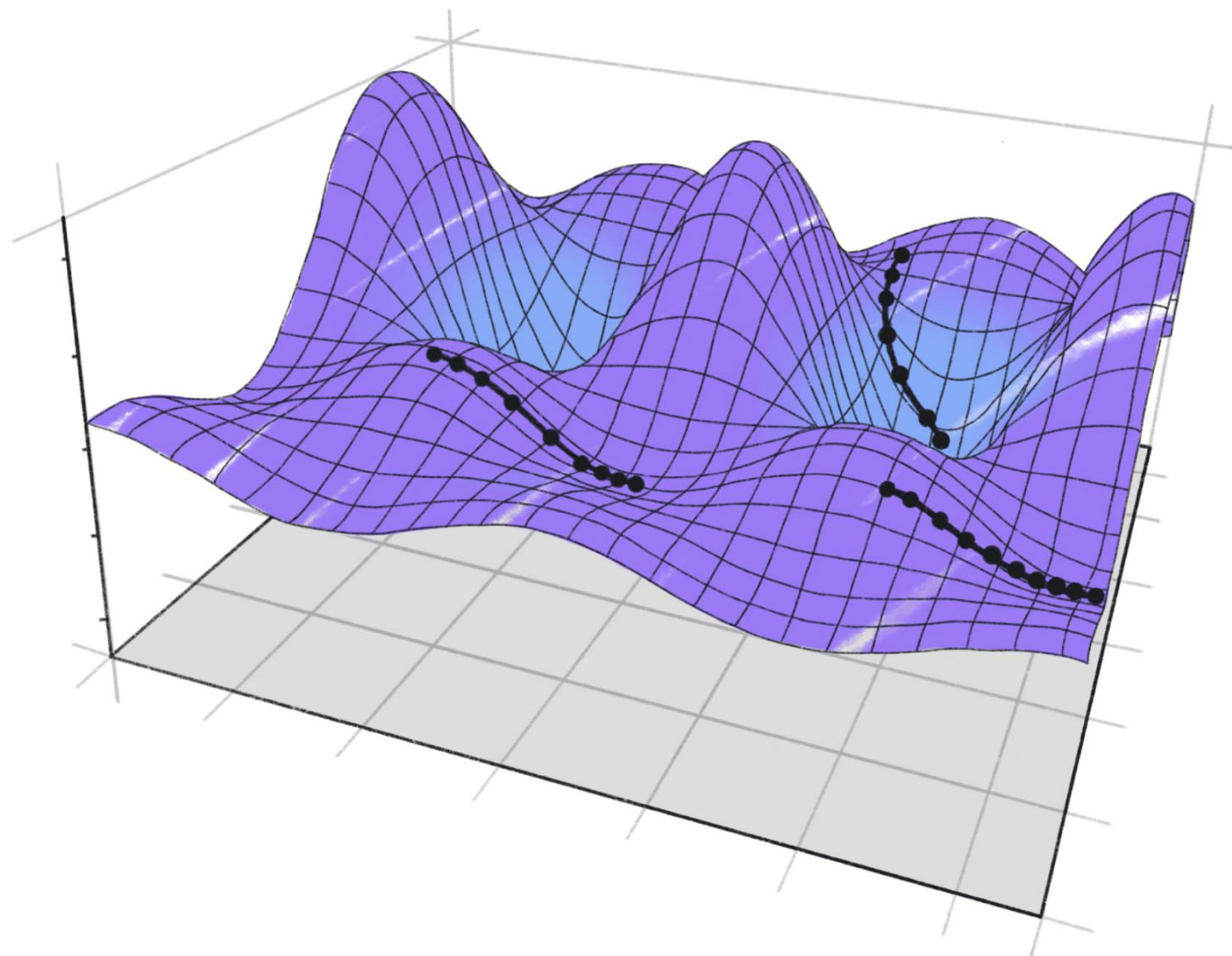
[ZZFeatureMap](#)

- Can be seen as the input state of the circuit.
- The choice of the reference state can impact the convergence of the algorithm (Optimization loop).
- Can be the default state! (The all 0 state).
- In the machine learning case it must also include parameterized gates to account for the different input data.

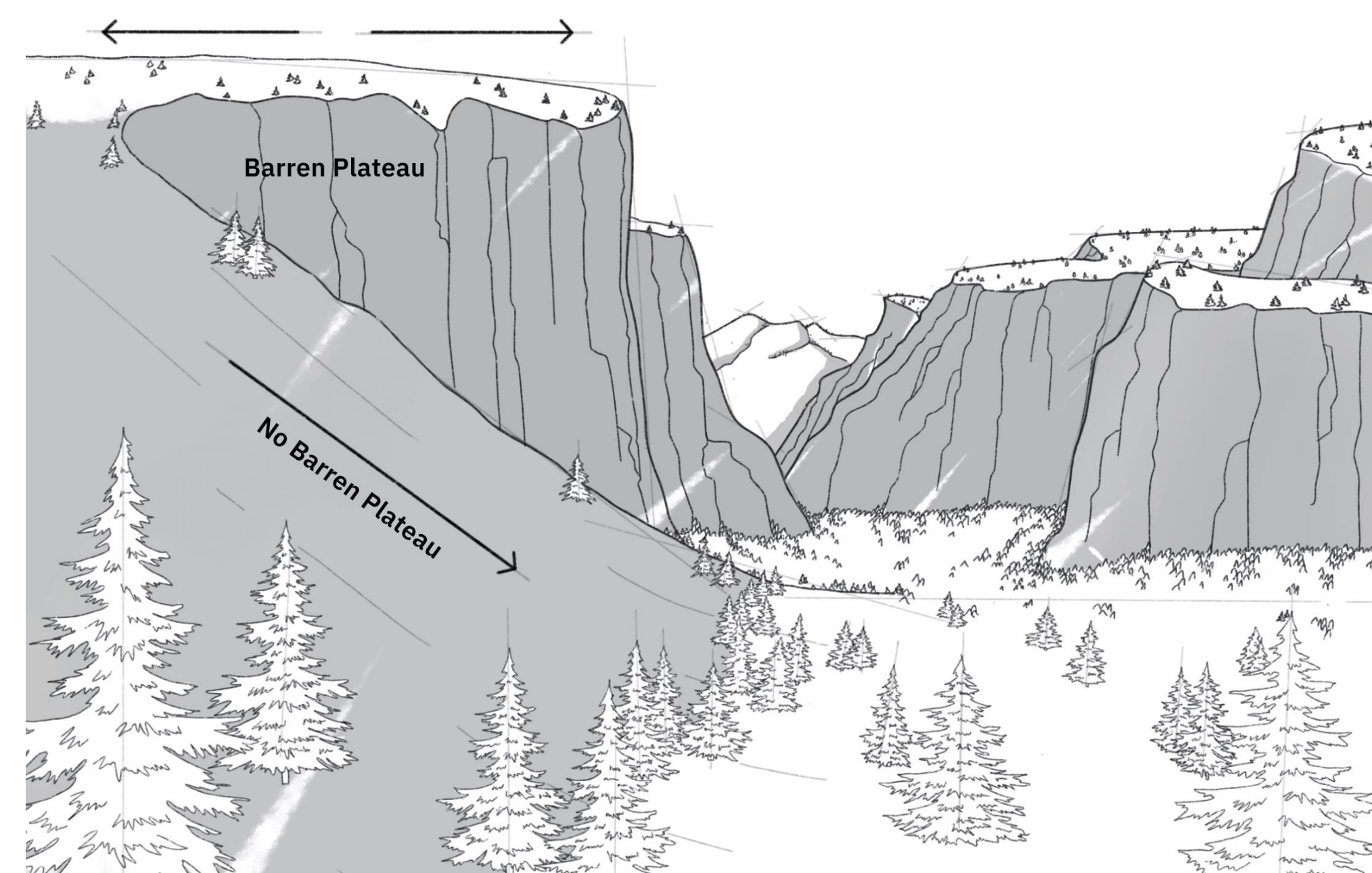
Optimization loop



- Any classical optimization function can be used, gradient based or gradient free.
- The quantum circuit can be seen as a “black box” which is optimized.
- We need to take care to not land in a local minimum instead of the global.



Barren plateau



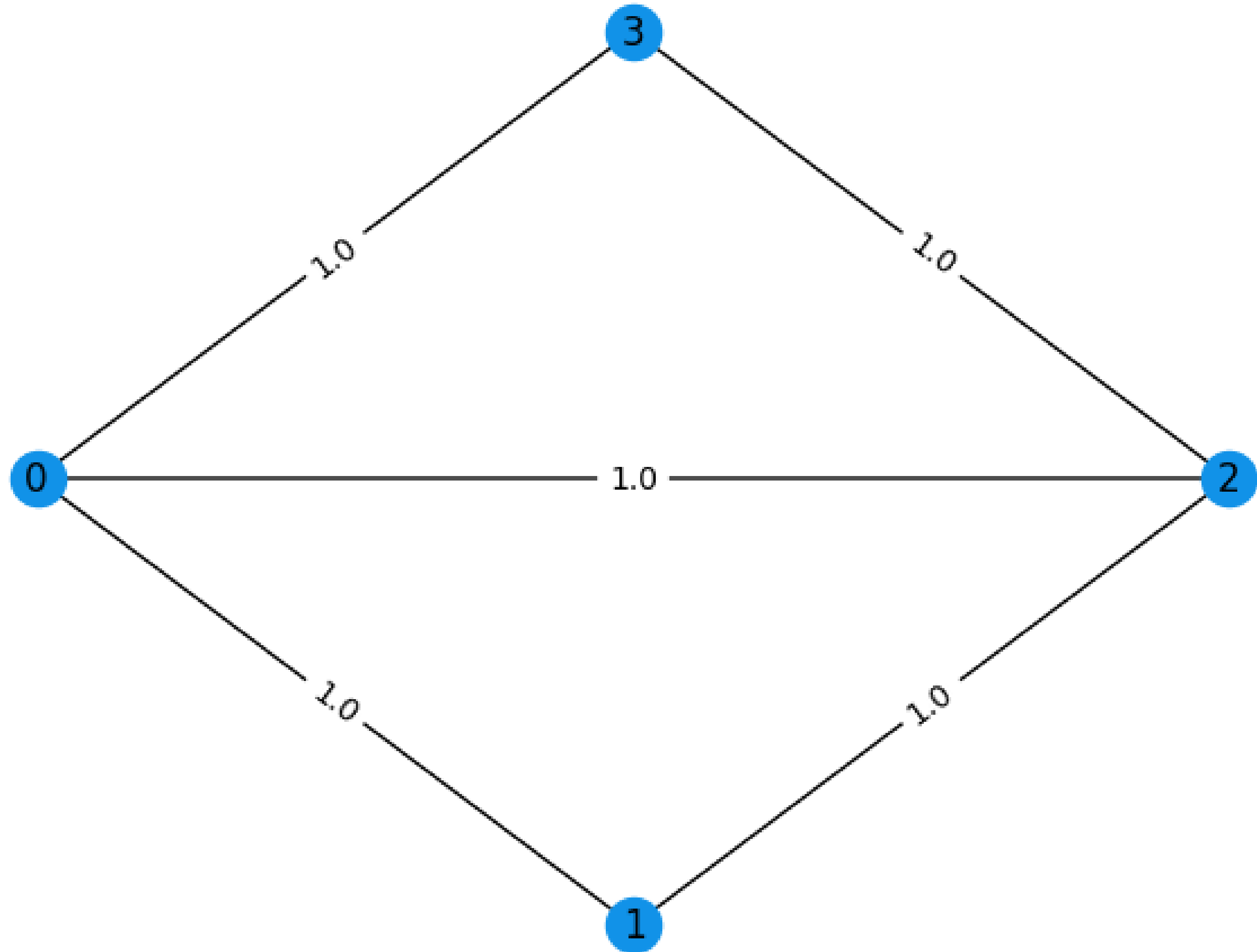
- The Barren Plateau occurs during the optimization loop if the cost function reaches a local plateau (flat area).
- The existence of Barren Plateaus makes it hard to find the global minimum of the function.
- Still an active research topic, there is no guaranteed way to prevent it, however, there are best practices (warm states, local cost functions).

Overview

- ✓ Recap Limitations and Variational Algorithms:
 - Limitations of Quantum Computing
 - Why should we use Variational Algorithms?
- ✓ • Variational Algorithms
 - Concept
 - Ansatz
 - Reference State
 - Cost Functions
 - Optimization Loops
- QAOA: Variational algorithms for optimization
 - QUBO
 - Adiabatic theorem
 - Hands-on
- VQE: Variational algorithms for ground energy estimation
 - Variational principle
 - Hands-on

Quantum Approximate Optimization Algorithm (QAOA):

Variational Algorithms for Optimization



Why is optimization important

Business Imperative

Need:

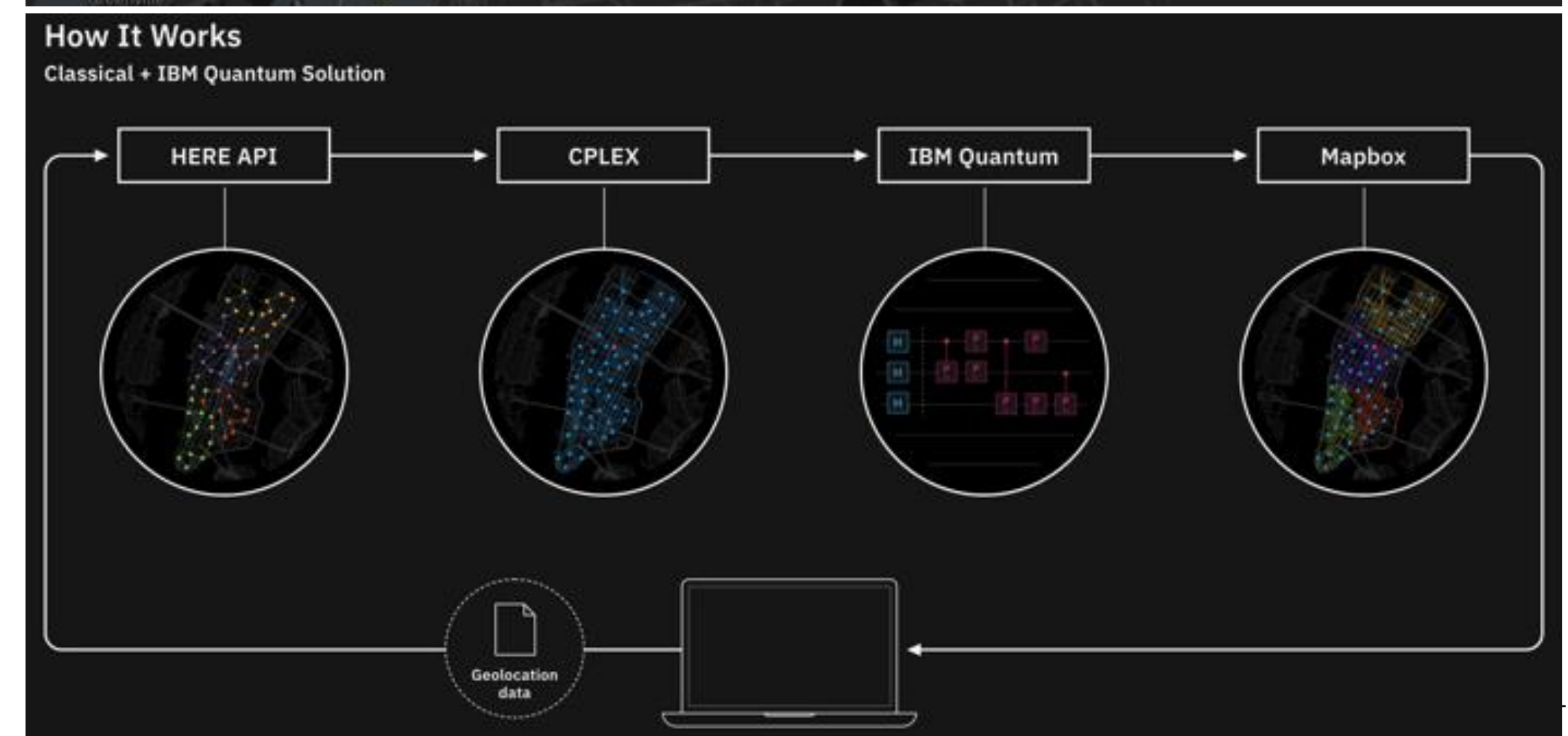
Routing optimization is a critical problem for many industries such as logistics, supply chain, automotive, telecommunications, and manufacturing.

Value:

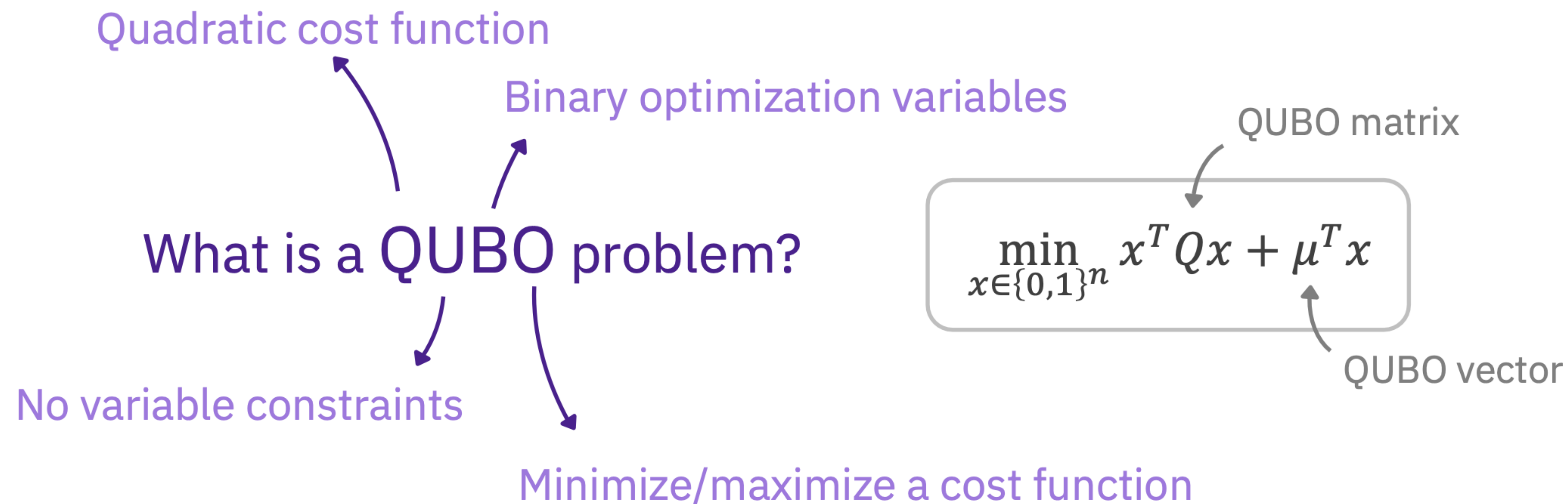
Reduces the total cost of delivery:

- Vehicle cost per day
- Driving cost per hour
- Distance cost per mile/km

Constraints: 1,200 delivery locations, vehicle capacity, single depot and delivery time windows



Target problem for QAOA: Quadratic Unconstrained Binary Optimization QUBO



This cost function can be encoded into a cost Hamiltonian H_C

How can we do Optimization?

How can we solve QUBOs on quantum computers?

- Quantum computers require for us to formulate our problems as **Hamiltonian problems**.
- Hamiltonian: mathematical description of a physical system's **energy**, in terms of operators or matrices. For an eigenstate $|x\rangle$ of the system,

$$H|x\rangle = E_{|x\rangle}|x\rangle \quad E_{|x\rangle} = \langle x|H|x\rangle$$

- *Goal*: to find a Hamiltonian, H_C , which **encodes our cost function**, $C(x)$.

$$H_C|x\rangle = C(x)|x\rangle$$

- If we can encode $C(x)$ in some H_C , we can then minimize it by finding the system's lowest energy state, i.e. the system's **ground state**

$$|x_{\text{optimal}}\rangle = \operatorname{argmin}_{|x\rangle \in \mathcal{H}} E_{|x\rangle}$$

So, how do we find H_C ?

1. Map each of the optimization variables to a qubit using the substitution

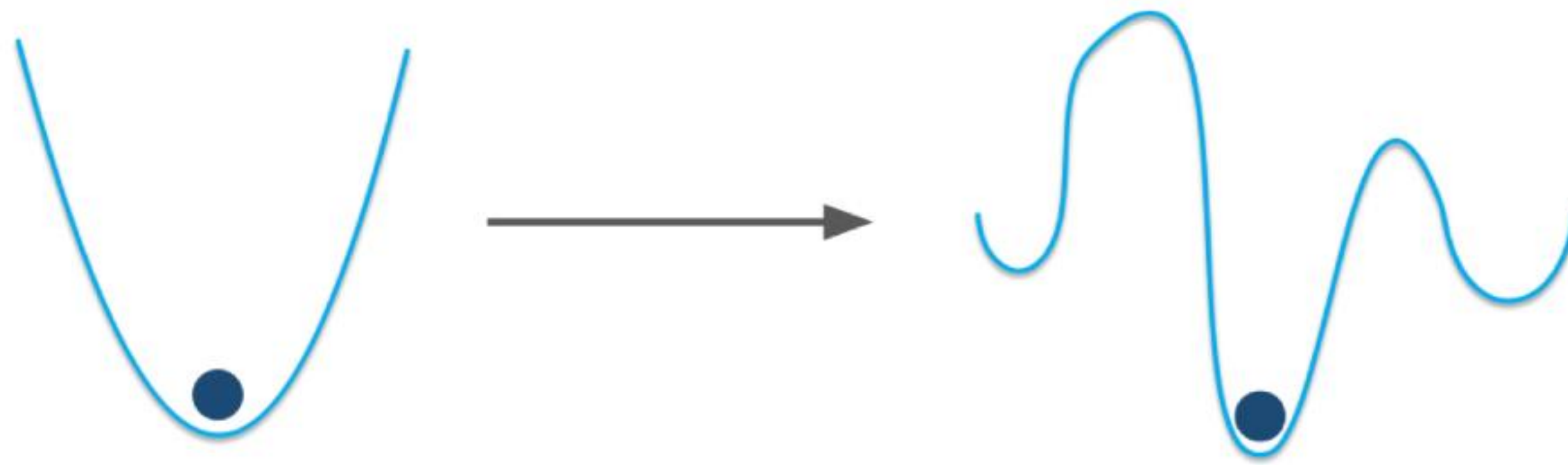
$$x_i = \frac{1 - z_i}{2}, \quad z_i \in \{-1, 1\}$$

2. Promote z_i to a Pauli spin operator Z_i , where $Z_i |x_i\rangle = (-1)^{x_i} |x_i\rangle$. This is equivalent to the above since $(-1)^{x_i} = 1 - 2x_i$ for $x_i \in \{0, 1\}$.

3. Make the substitution,

$$\begin{aligned}
 H_C &= \sum_{ij} Q_{ij} \frac{1 - Z_i}{2} \frac{1 - Z_j}{2} + \sum_i \mu_i \frac{1 - Z_i}{2} = \sum_{ij} \frac{Q_{ij}}{4} - \sum_i \left(\sum_j \frac{Q_{ij}}{4} \right) Z_i - \overbrace{\sum_j \left(\sum_i \frac{Q_{ij}}{4} \right) Z_j}^{Q_{ij} \text{ symmetric } i \leftrightarrow j} + \sum_{ij} \frac{Q_{ij}}{4} Z_i Z_j + \sum_i \frac{\mu_i}{2} - \sum_i \frac{\mu_i}{2} Z_i = \\
 &= \underbrace{\sum_{ij} \frac{Q_{ij}}{4} Z_i Z_j - \sum_i \frac{1}{2} \left(\mu_i + \sum_j Q_{ij} \right) Z_i + \sum_{ij} \frac{Q_{ij}}{4} + \sum_i \frac{\mu_i}{2}}_{\text{This is an Ising Hamiltonian}}
 \end{aligned}$$

Adiabatic theorem, intuition behind QAOA



Adiabatic theorem: If a quantum system starts in the ground state of a time-dependent Hamiltonian $H(t)$ and the Hamiltonian changes sufficiently slowly, the system will remain in the ground state throughout the evolution.

We define:

$$H(t) = (1 - t)H_M + tH_C, t \in [0,1]$$

- H_M : mixer Hamiltonian whose ground state we know
- H_C : cost Hamiltonian whose ground state we seek

Time evolution:

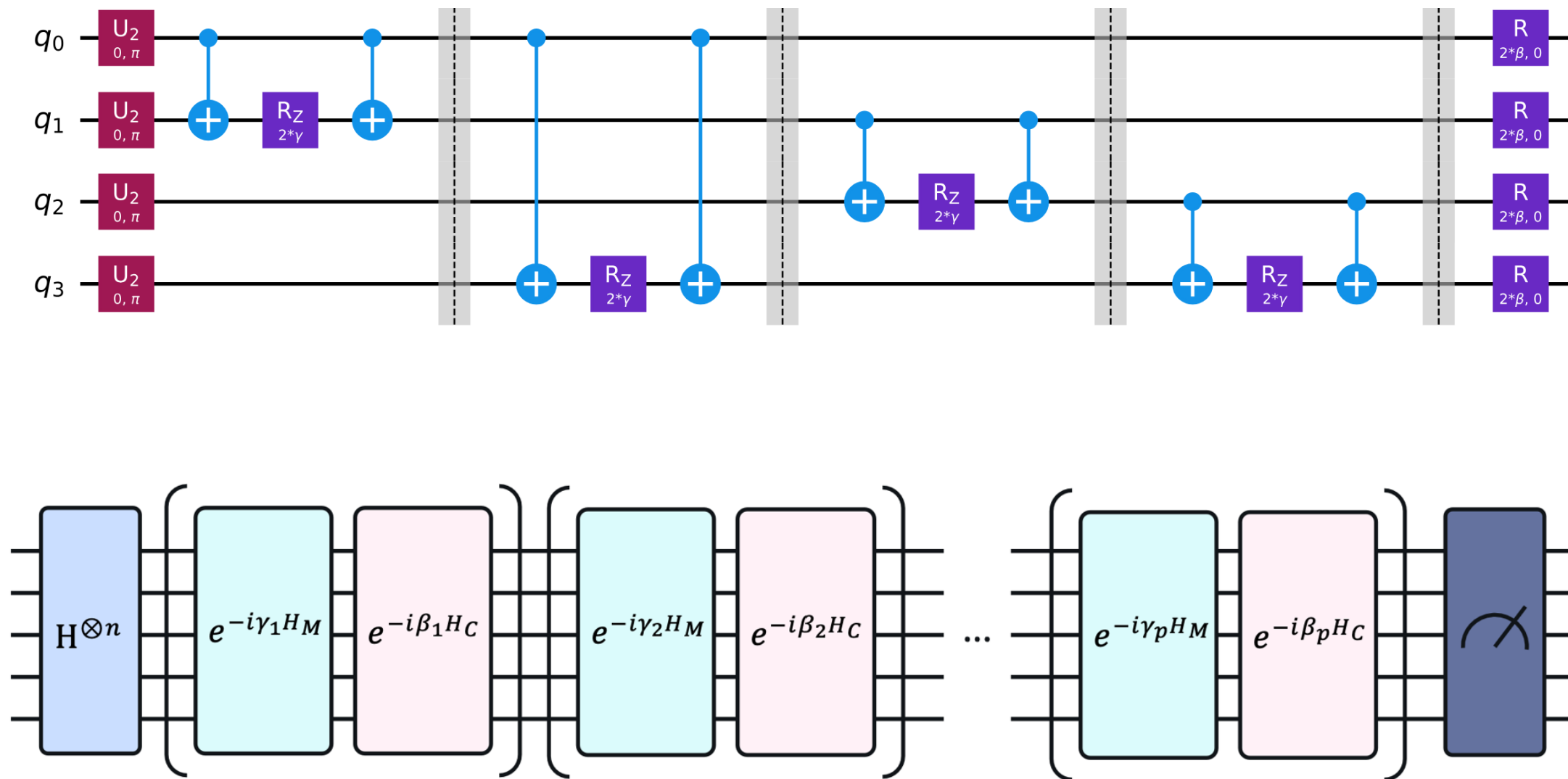
$$U(H, t, r) \approx \prod_{k=0}^{r-1} e^{-i(1 - k t / r) H_M t / r} e^{-i(k t / r) H_C t / r}$$

Variational parameters of QAOA:

$$\alpha_k = (1 - k t / r) t / r$$

$$\gamma_k = (k t / r) t / r$$

QAOA (Quantum Approximate Optimization Algorithm) Workflow



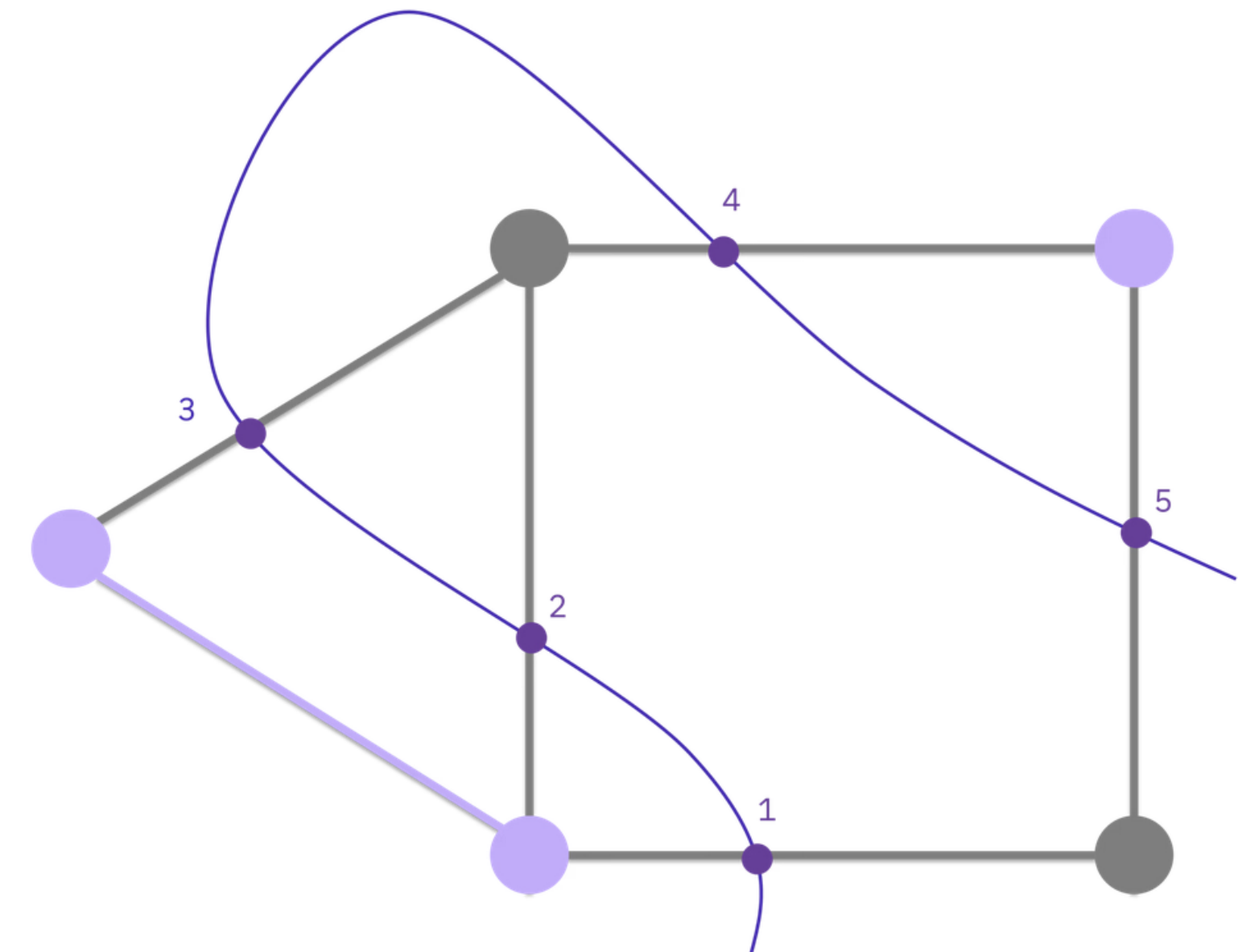
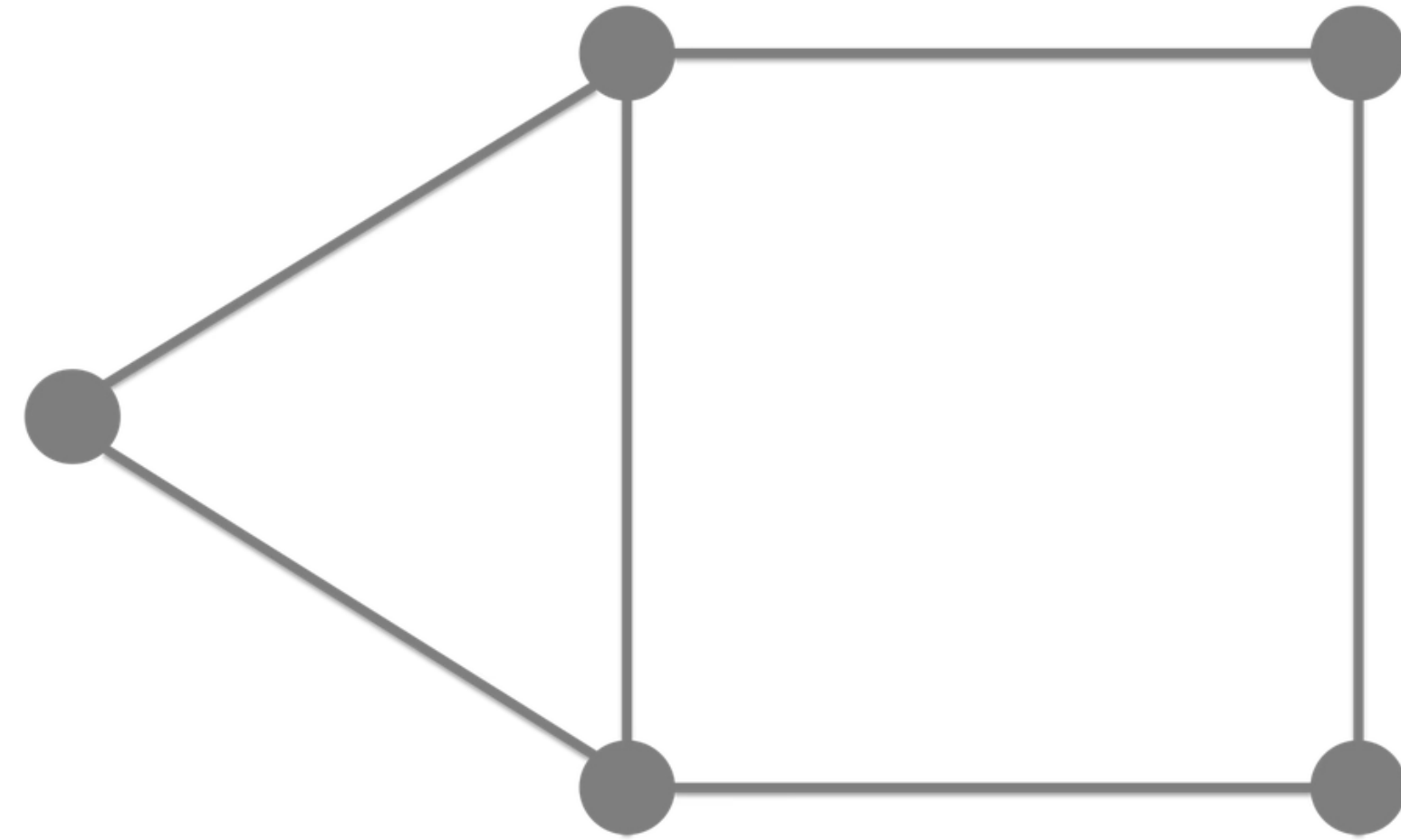
1. Map the real world application to an optimization problem (QUBO).
2. Map the QUBO problem to an (Ising) Hamiltonian which encodes the cost function.
3. Solve the ground state problem (using QAOA).

Qiskit Addon: Optimization Mapper

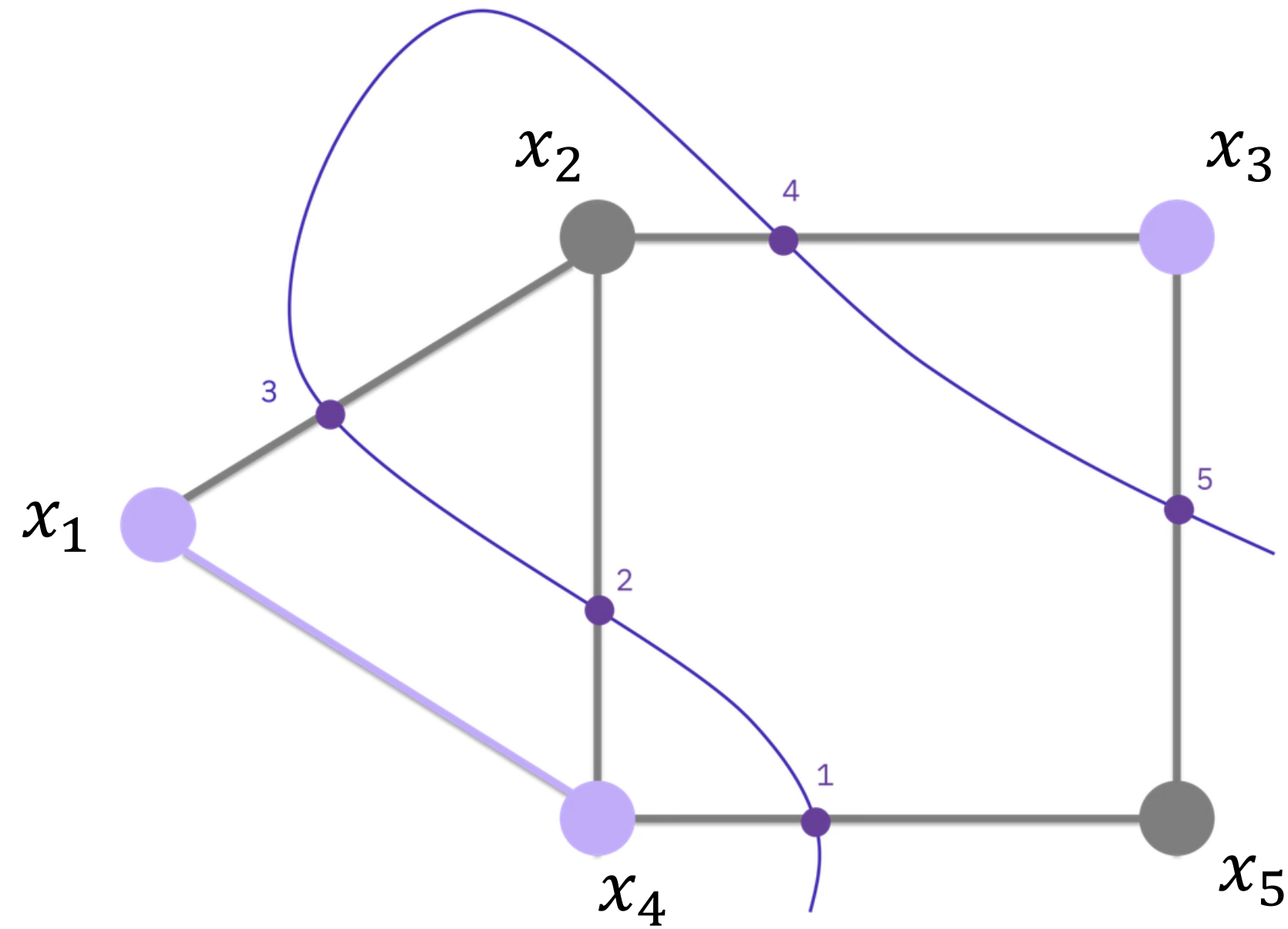
<https://qiskit.github.io/qiskit-addon-opt-mapper>

Hands on: Max-Cut

- Graph problem we want to maximize number of connections cut, by separating the graph in 2 parts.
- Is equivalent to minimizing the (Ising) Hamiltonian of a spin model.
- Can be naturally applied to binary classification (a known use case for quantum machine learning) or network design, clustering and more.



Explanation Max Cut Problem



$$x = [x_1, x_2, x_3, x_4, x_5]$$

$$x = [0, 1, 0, 0, 1]$$

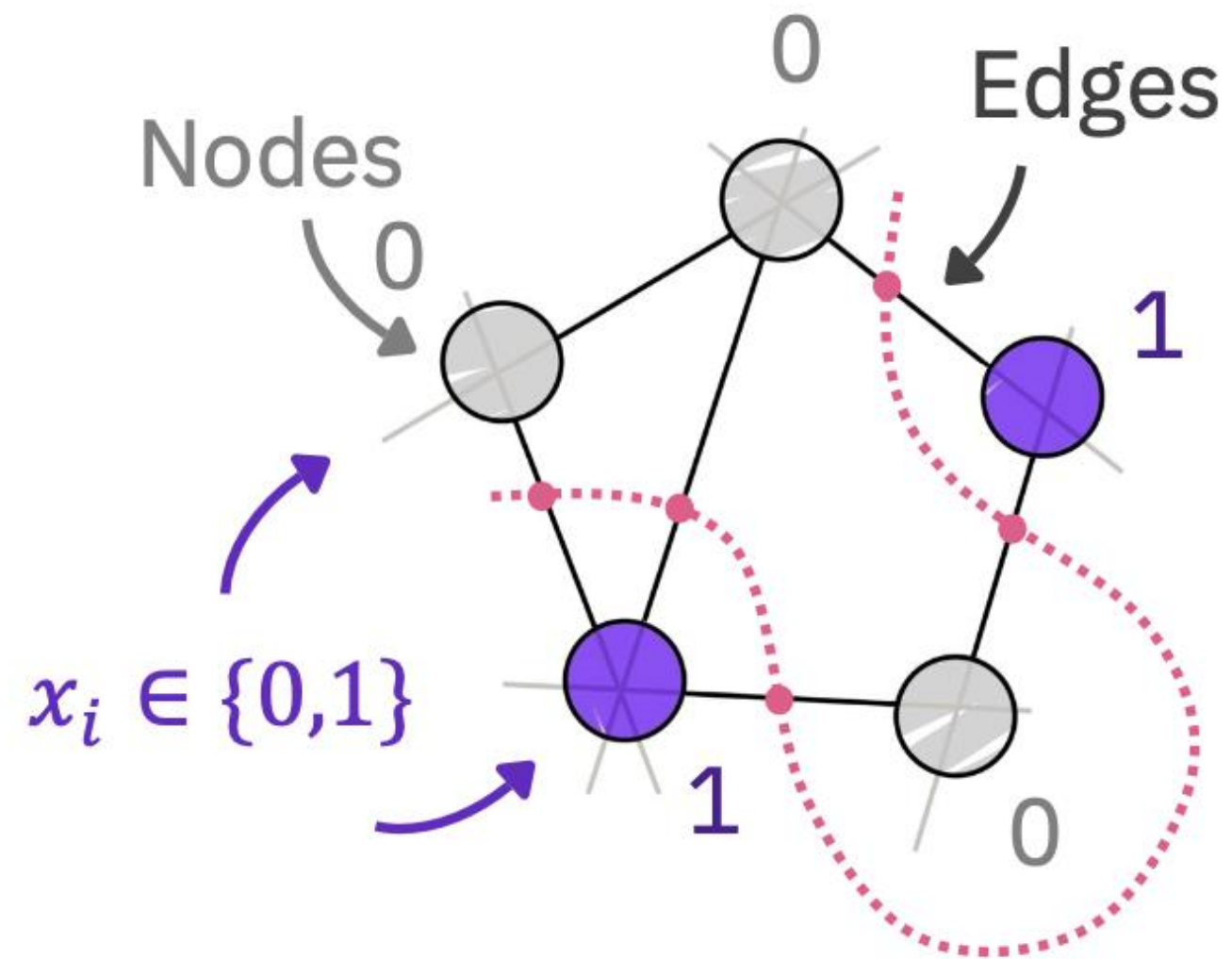
QUBO

$$\min_{x \in \{0,1\}^n} x^T Q x + \mu^T x$$



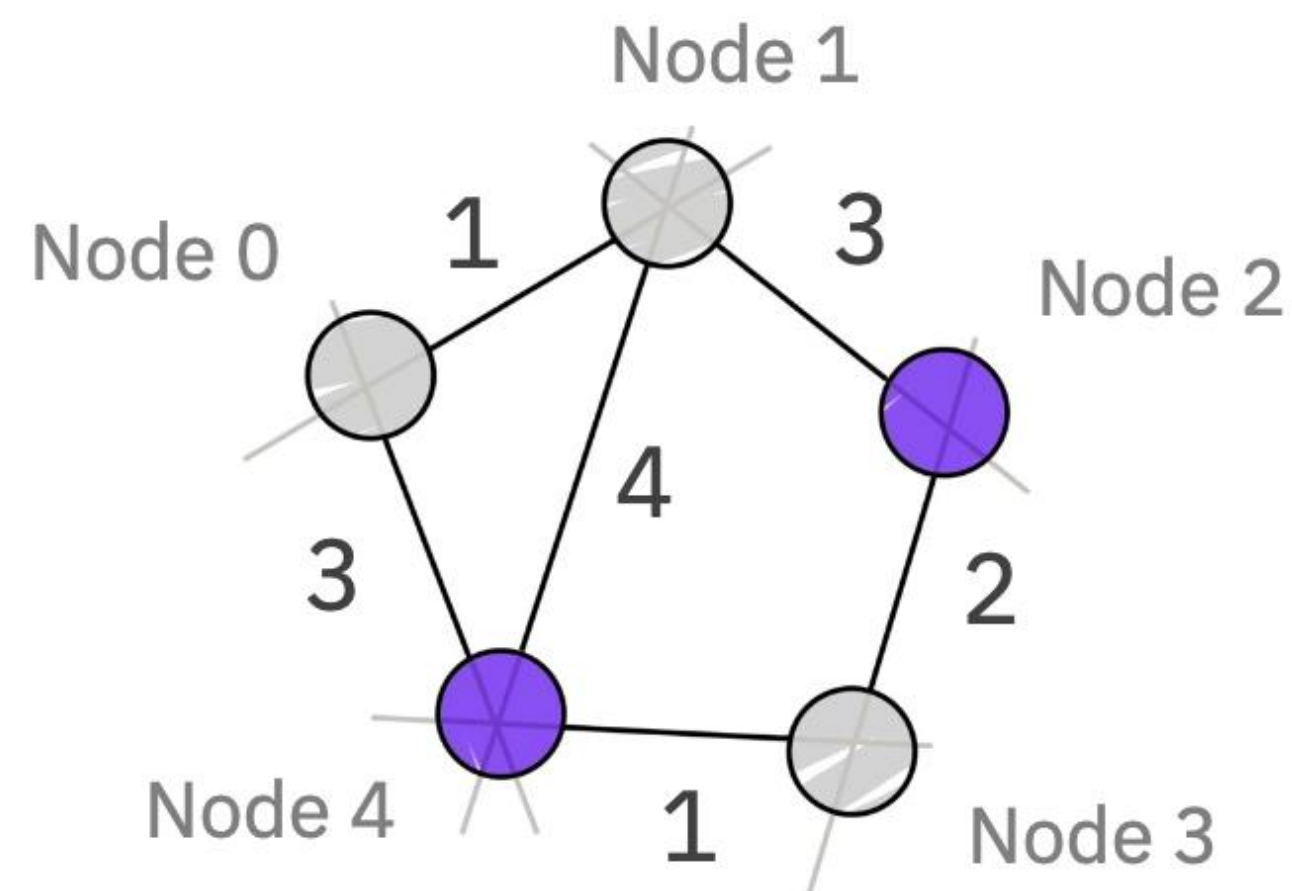
$$\min_{x \in \{0,1\}^n} \sum_{(i,j)} 2x_i x_j - x_i - x_j$$

Max-Cut: an example of QUBO



Goal: to partition nodes into two groups such that the number of edges connecting nodes from the different groups is maximized

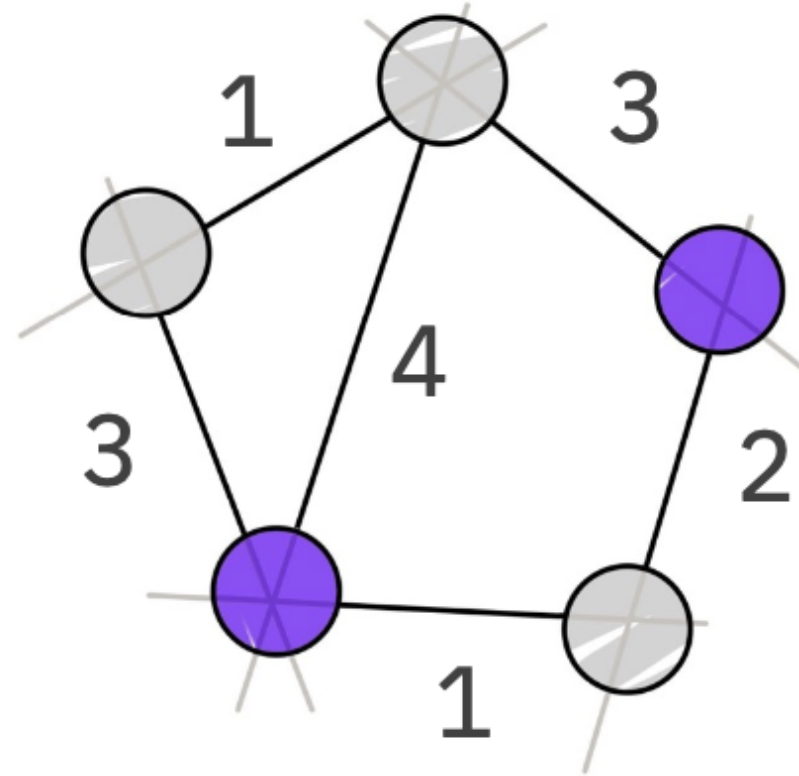
⇒ Leads to *binary* optimization variables, since we can assign each node a value of 0 or 1 depending on the group which they are on



We can also assign weights to the edges, which will give us our QUBO matrix, e.g.

$$w_{ij} = \begin{pmatrix} 0 & 1 & 0 & 0 & 3 \\ 1 & 0 & 3 & 0 & 4 \\ 0 & 3 & 0 & 2 & 0 \\ 0 & 0 & 2 & 0 & 1 \\ 3 & 4 & 0 & 1 & 0 \end{pmatrix}$$

Max-Cut: an example of QUBO



$$w_{ij} = \begin{pmatrix} 0 & 1 & 0 & 0 & 3 \\ 1 & 0 & 3 & 0 & 4 \\ 0 & 3 & 0 & 2 & 0 \\ 0 & 0 & 2 & 0 & 1 \\ 3 & 4 & 0 & 1 & 0 \end{pmatrix}$$



$$\min_{x \in \{0,1\}^n} x^T Q x + \mu^T x$$

Only edges between nodes of
different groups contribute



$$\begin{aligned} C(x) &= \sum_{i,j=1}^n w_{ij} x_i (1 - x_j) \\ &= \sum_{i,j=1}^n w_{ij} x_i - \sum_{i,j=1}^n x_i w_{ij} x_j \end{aligned}$$



$$\mu_i = \sum_{j=1}^n w_{ij}$$

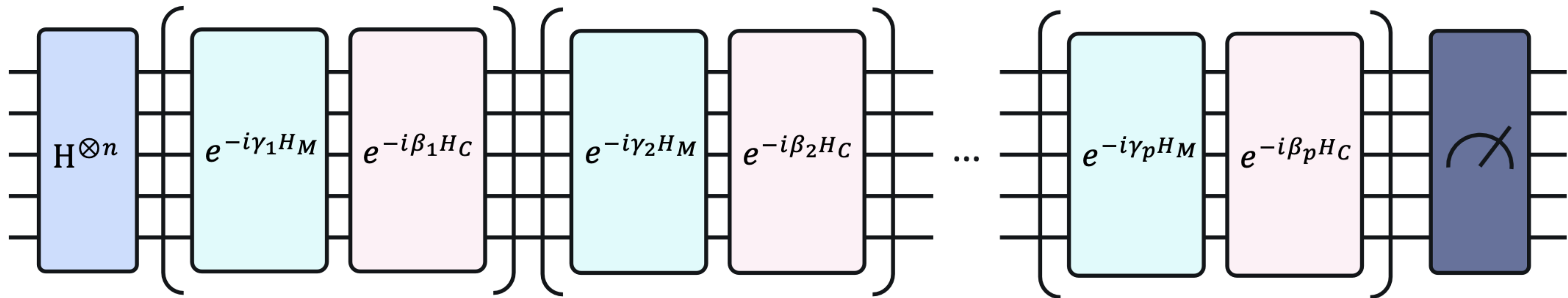
$$Q_{ij} = -w_{ij}$$



$$\begin{aligned} C(x) &= \sum_{i,j=1}^n x_i Q_{ij} x_j + \sum_{i=1}^n \mu_i x_i \\ &= x^T Q x + \mu^T x \end{aligned}$$

The quantum circuit

1. Define $H_M = -\sum_i X_i$ and H_C such that it encodes a QUBO
2. Apply Hadamard gates to all qubits to begin in the ground state of H_M , i.e. $|+\rangle^{\otimes n}$
3. Apply p layers, each consisting of two terms, $e^{-i\gamma_k H_M}$ and $e^{-i\beta_k H_C}$ for $k = 1, \dots, p$
4. Measure

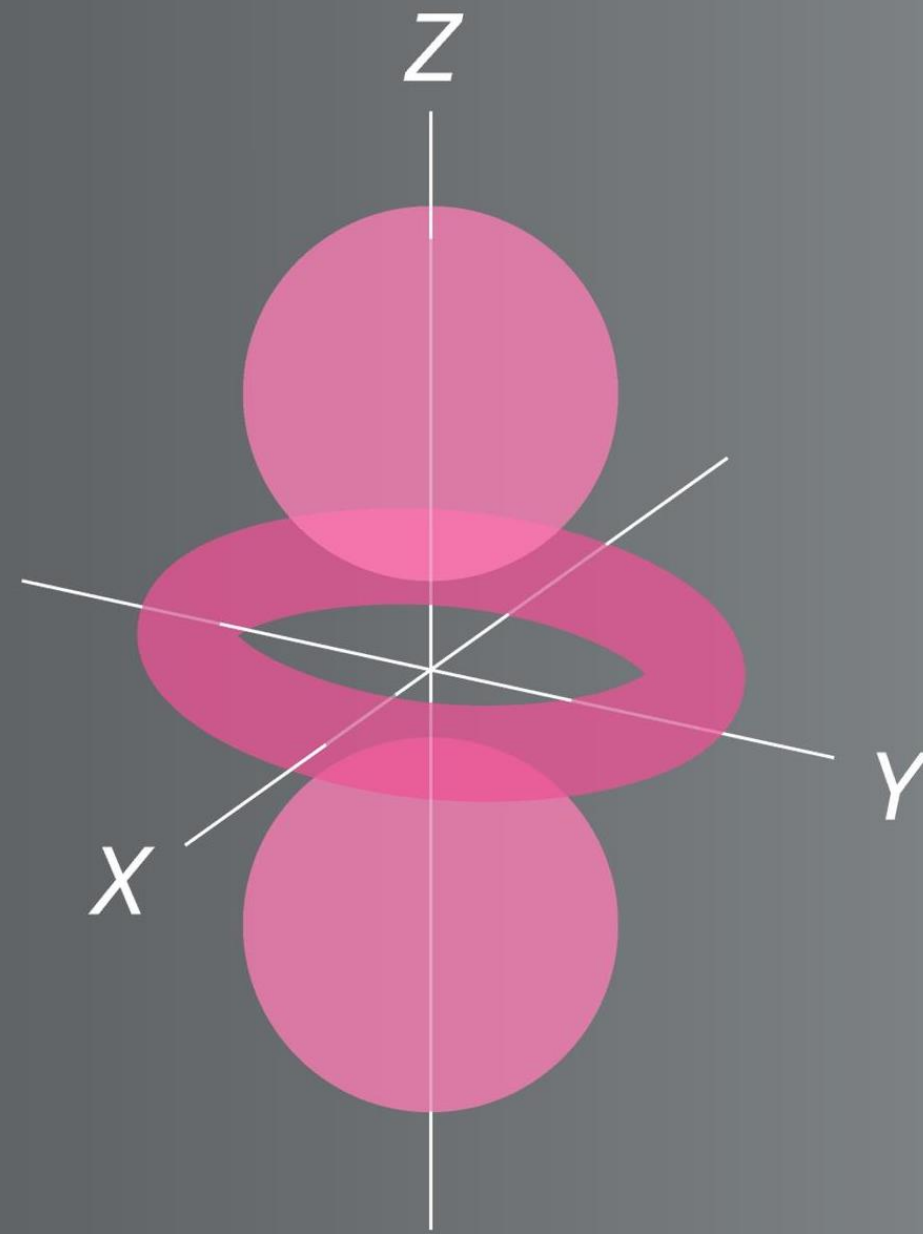


Overview

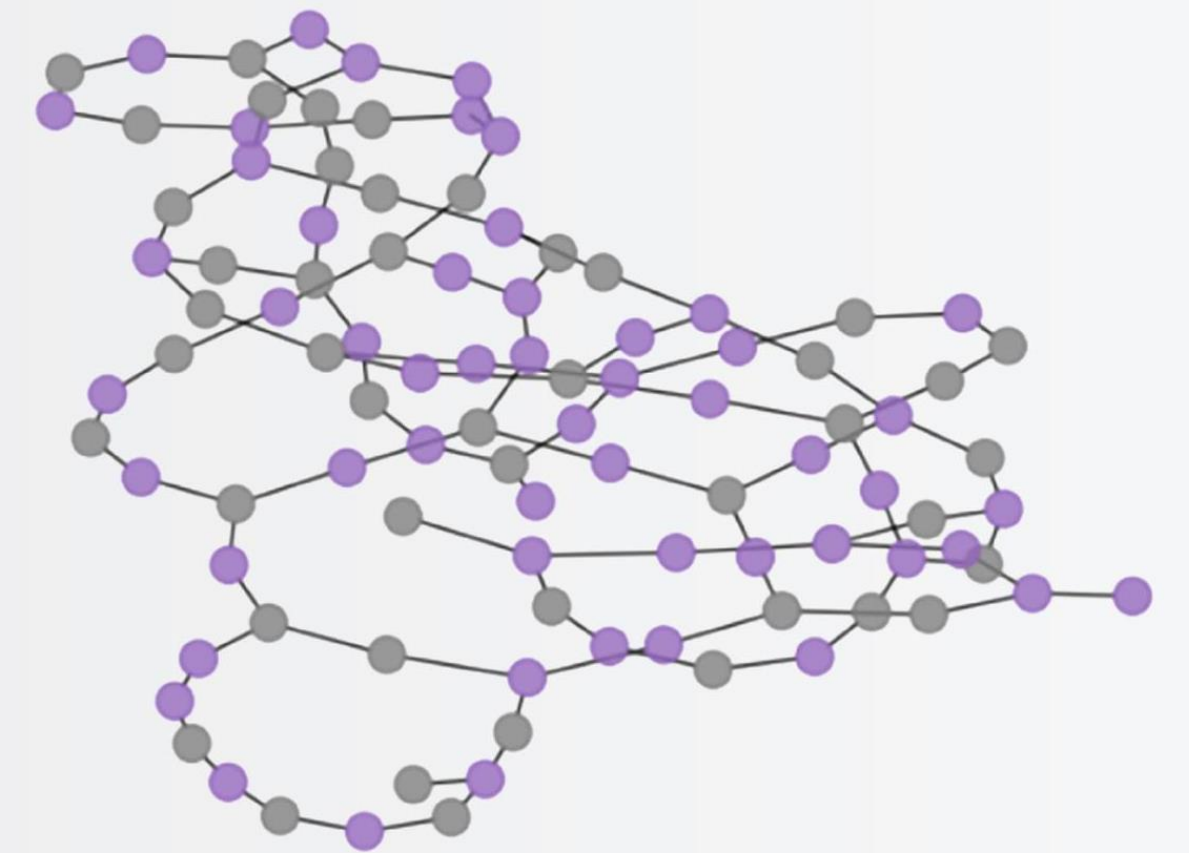
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 - Variational principle
 - Recipe

Variational Quantum Eigensolver (VQE):

Variational Algorithms for Ground Energy Estimation



$$H|\psi_0\rangle = E_0|\psi_0\rangle$$



$$H|\psi_{\text{opt}}\rangle = c_{\text{opt}}|\psi_{\text{opt}}\rangle$$

1. VQE finds the **ground state** of a given Hamiltonian $\hat{H}$, usually represented as a linear combination of Pauli terms $\hat{P}_j$
2. Based on Rayleigh-Ritz variational principle:

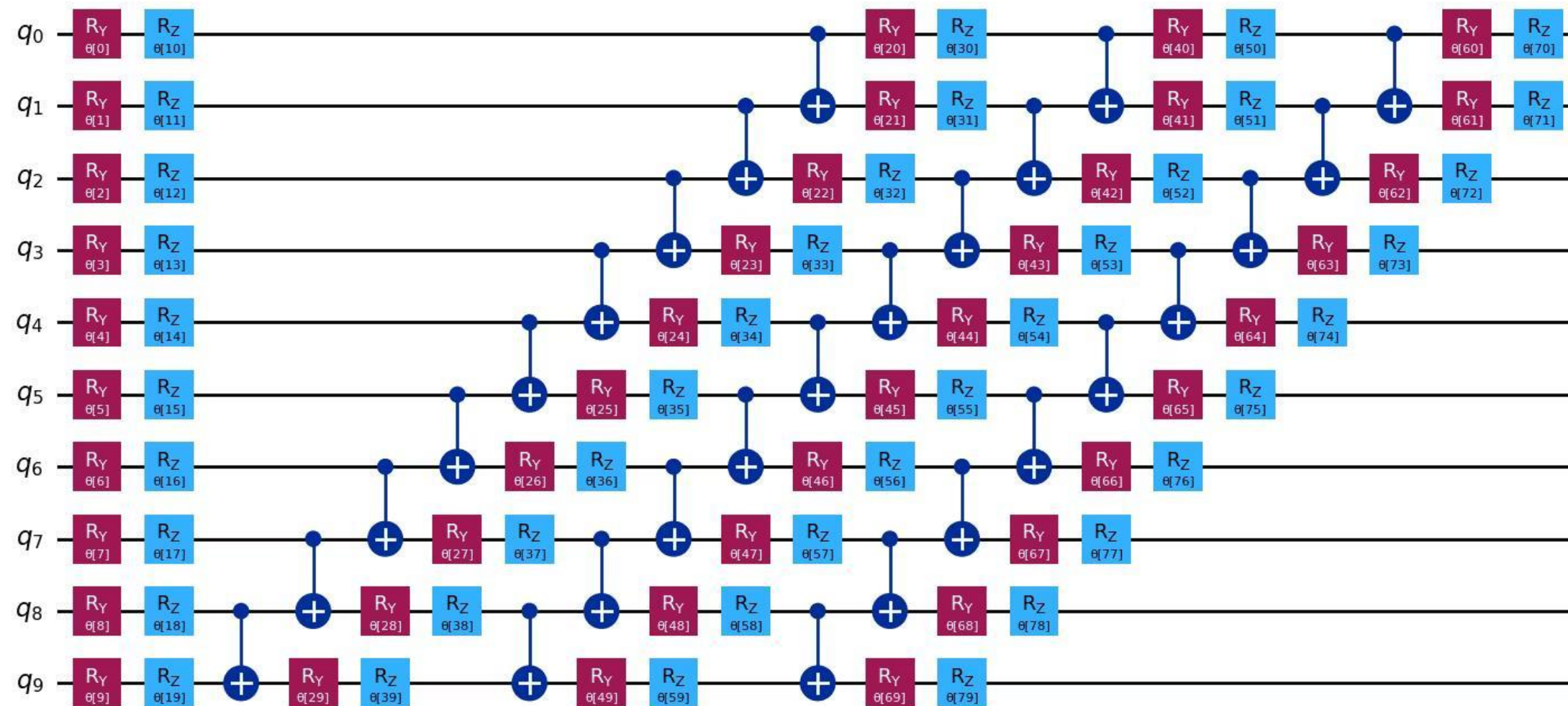
$$\hat{H} = \sum_i E_i |\Psi_i\rangle\langle\Psi_i|$$

$$E_0 \leq \frac{\langle\Psi|\hat{H}|\Psi\rangle}{\langle\Psi|\Psi\rangle}$$

for any trial state $|\Psi\rangle$. When the trial state equals the ground state, i.e. $|\Psi\rangle = |\Psi_0\rangle$:

$$E_0 = \frac{\langle\Psi_0|\hat{H}|\Psi_0\rangle}{\langle\Psi_0|\Psi_0\rangle}$$

VQE (Variational Quantum Eigensolver) Workflow



1. Hamiltonian Encoding:

Represent the system's Hamiltonian $\hat{H}$ as a weighted sum of Pauli terms:

$$\hat{H} = \sum_j c_j \hat{P}_j$$

where $\hat{P}_j$ are tensor products of Pauli matrices and $c_j \in \mathbb{R}$.

2. Ansatz Selection:

Choose a parameterized quantum circuit $U(\vec{\theta})$ to generate a family of trial states:

$$|\Psi(\vec{\theta})\rangle = U(\vec{\theta})|0\rangle$$

3. Qiskit Runtime estimator:

Estimate the expectation value of the Hamiltonian:

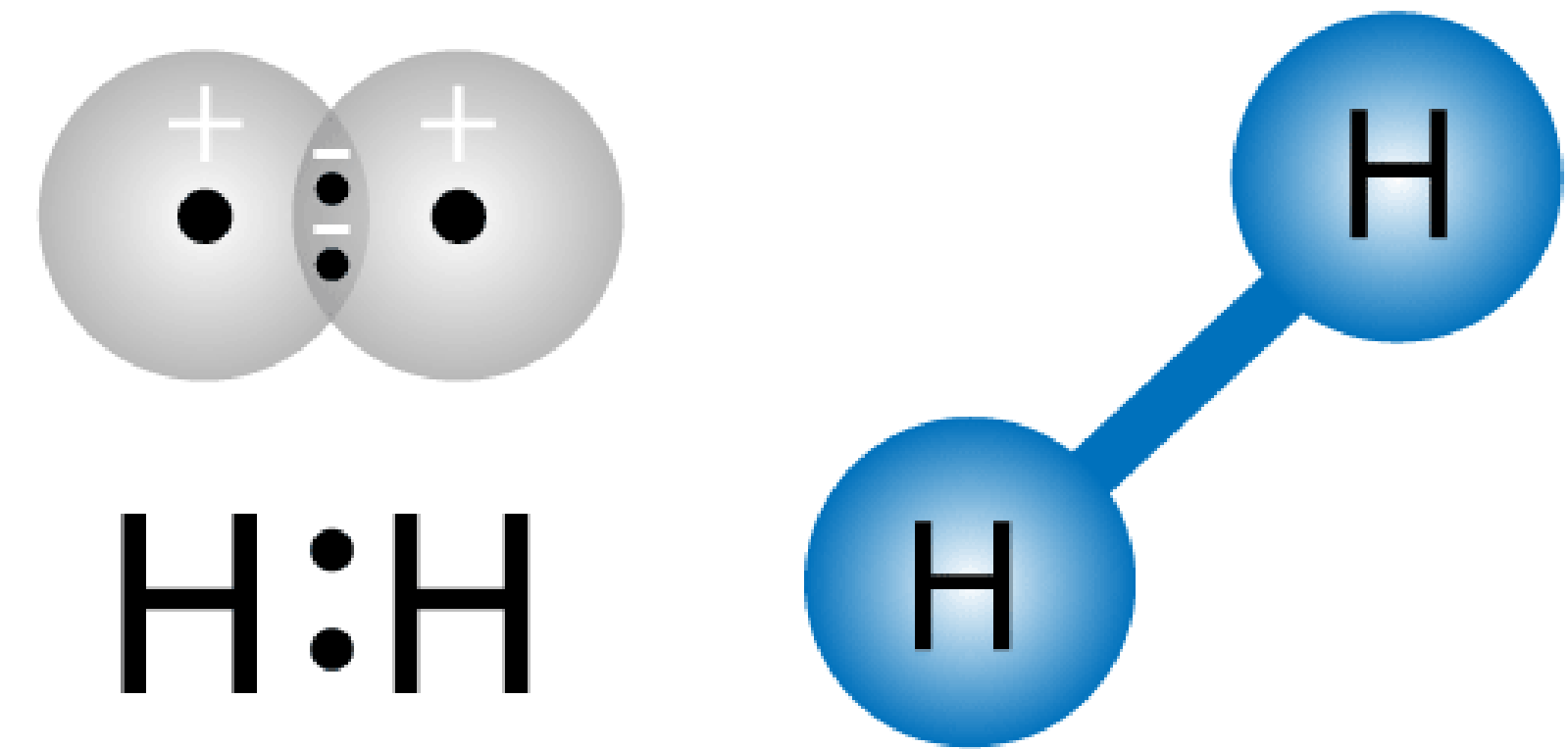
$$E(\vec{\theta}) = \sum_j c_j \langle \Psi(\vec{\theta}) | \hat{P}_j | \Psi(\vec{\theta}) \rangle$$

4. Classical Optimization:

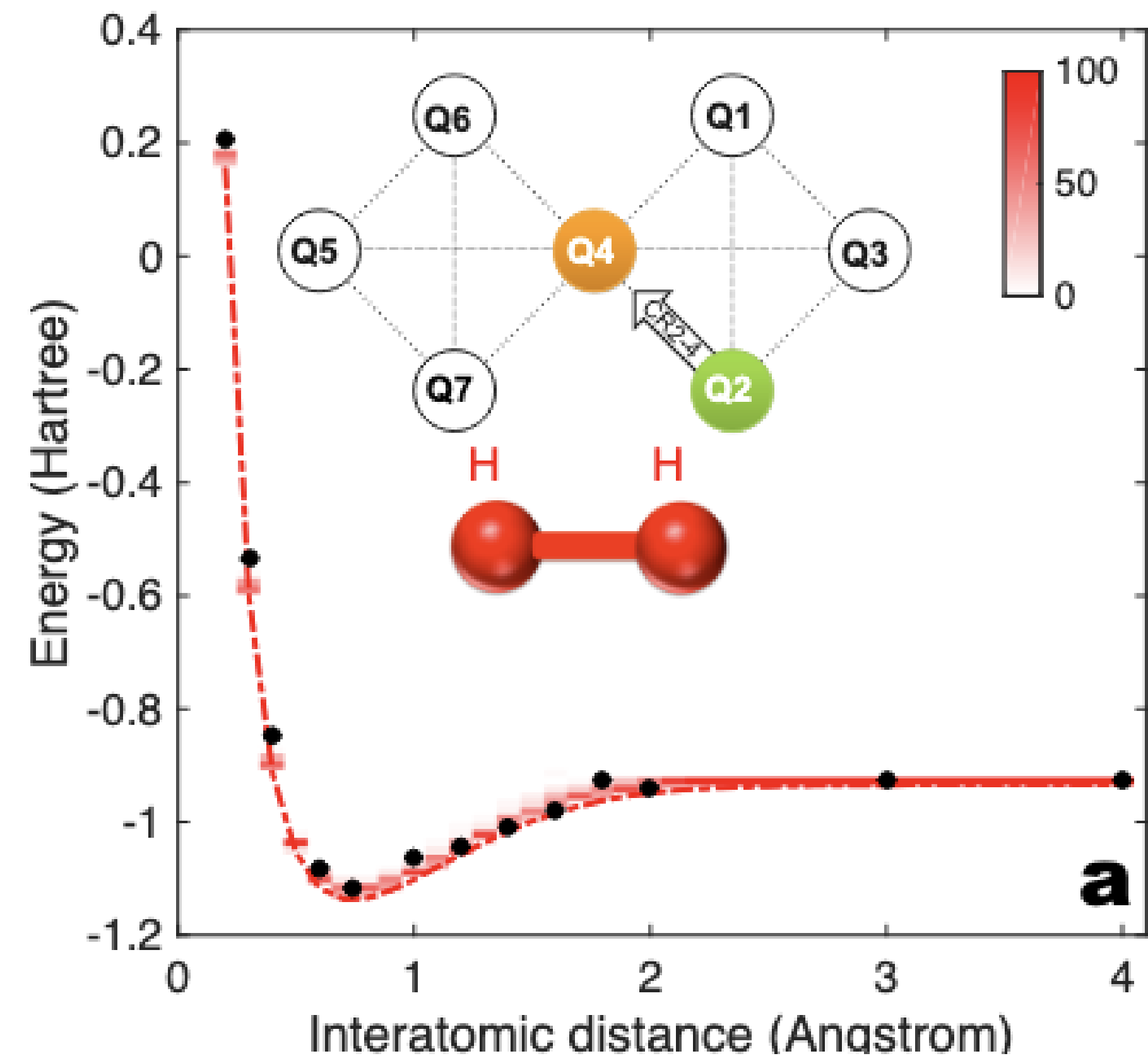
Use a classical optimizer to update $\vec{\theta}$ to minimize $E(\vec{\theta})$ and accomplish

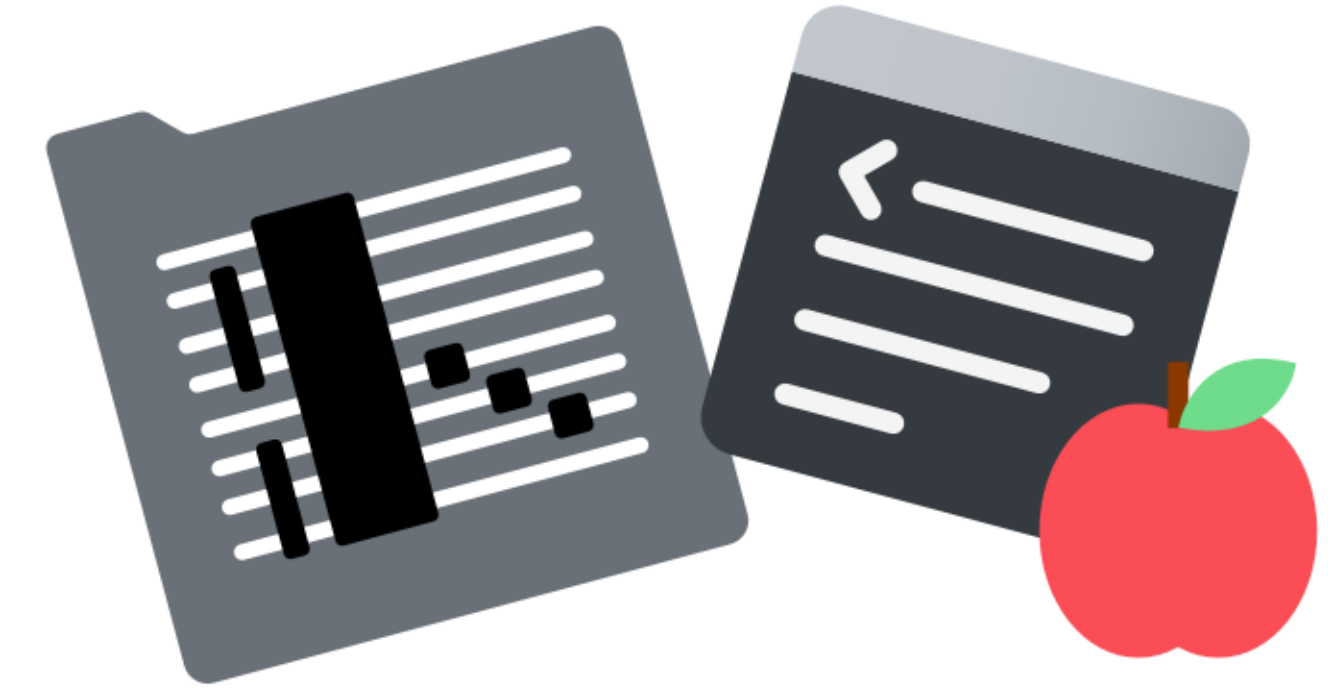
$$|\Psi(\vec{\theta})\rangle \approx |\Psi_0\rangle$$

Hands on: Ground state estimation of H_2 molecule



Hydrogen (H_2)





Further reading

Courses:

Variational algorithm design:

<https://quantum.cloud.ibm.com/learning/en/courses/variational-algorithm-design>

Utility scale quantum computing:

<https://quantum.cloud.ibm.com/learning/en/courses/utility-scale-quantum-computing>

Quantum chemistry with VQE:

<https://quantum.cloud.ibm.com/learning/en/courses/quantum-chem-with-vqe>

Tutorials:

QAOA:

<https://quantum.cloud.ibm.com/docs/en/tutorials/quantum-approximate-optimization-algorithm>

Ground-state energy estimation of the Heisenberg chain with VQE:

<https://quantum.cloud.ibm.com/docs/en/tutorials/spin-chain-vqe>

Qiskit Addon: Optimization Mapper

<https://qiskit.github.io/qiskit-addon-opt-mapper>